
Girsanov-based Privacy in Stochastic Gradient Langevin Dynamics for Bayesian Federated Learning

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Abstract

Langevin-based methods such as stochastic gradient Langevin dynamics (SGLD) are useful tools for scalable Bayesian learning, including in distributed and federated settings where privacy is a primary design objective. However, their privacy guarantees are still mostly understood through stepwise composition, which treats each iteration as an independent mechanism and obscures how the dynamics and subsampling structure amplify privacy. Recent diffusion-based analyses using Girsanov’s theorem yield uniform-in-time bounds, but either require full-batch gradients or ignore the role of minibatch and client sampling. In this work we revisit the Girsanov framework through the lens of Rényi differential privacy and concentration inequalities, focusing on the practically relevant “final iterate only” release. In the centralized setting, we show that minibatch SGLD with a nonconvex data term and a strongly convex regularizer enjoys uniform-in-time RDP, and that the resulting privacy loss scales linearly in the subsampling rate $q = s/m$ for small q , recovering privacy amplification by minibatching within a diffusion-based analysis. We then extend our approach to distributed SGLD with an arbitrary hospital scheduler and to federated SGLD with surrogate-based conductive gradients, deriving analogous uniform-in-time RDP guarantees that capture amplification from both client sampling and local minibatching. Taken together, our results demonstrate that Girsanov-style, pathwise privacy arguments can accommodate stochastic gradients and heterogeneous data while capturing the privacy benefits inherent to standard federated learning design choices.

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1 Introduction

1.1 Bayesian federated learning

Pushing statistical learning to the edge of distributed networks exploits the computational capacity of client devices while preserving client privacy. Federated learning describes the process of training models directly on user-held devices and aggregating their updates on a central server. Federated learning has seen early adoption among major service providers [Li et al., 2020] with potential applications ranging from learning patterns of pedestrian movement for autonomous vehicle navigation to predicting risk of heart attack and stroke.

At each round of federated learning, a subset of clients, which we will refer to as “hospitals,”¹ performs local training on (non-i.i.d.) data, pushing gradient updates to the central server, which consolidates those updates and sends the new global model to another (randomly sampled) subset, and so on until convergence.

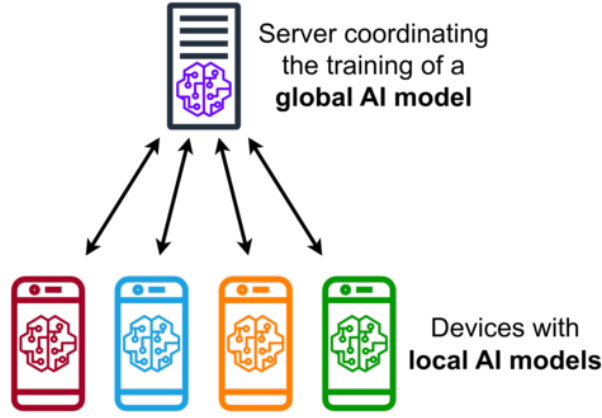


Figure 1: Centralized federated learning protocol. Image credit: Hrushikesh Bansal, Wikimedia Commons, CC BY-SA 4.0.

Given m total hospitals, the objective function of the server is

$$\min_{\theta} F(\theta), \quad F(\theta) := \sum_{k=1}^m p_k F_k(\theta), \quad (1)$$

with $p_k \geq 0$, $\sum_{k=1}^m p_k = 1$, and F_k the objective for the k^{th} hospital (e.g., empirical risk over the

hospital’s local data). It is common to choose $p_k = \frac{1}{n}$ or $p_k = \frac{n_k}{n}$, with $n = \sum_{k=1}^m n_k$.

The Bayesian learning paradigm offers a principled way to address several core challenges in federated learning. In particular, the tiny, highly non-IID datasets held by intermittently participating clients warrant calibrated uncertainty estimates, which Bayesian approaches naturally motivate. Bayesian methods also leverage prior knowledge of parameters to give tighter estimates in data-sparse settings [Cao et al., 2023].

Rather than learning a full posterior distribution in practice, one typically views the standard federated optimization objective through a Bayesian lens: the prior $p(\theta)$ corresponds to a regularizer, and the loss is interpreted as a negative log-likelihood term. This reframing provides a probabilistic interpretation of the learning problem and suggests ELBO-like formulations.

In a centralized Bayesian setting, the posterior would be

$$p(\theta_g | \mathcal{D}) \propto p(\theta_g) p(\mathcal{D} | \theta_g), \quad (2)$$

¹This corresponds to the illustrative use case of training a medical model across data from multiple hospitals.

but in federated learning the server does not have access to the full dataset $\mathcal{D} = \bigcup_{m=1}^M \mathcal{D}_m$. Instead, each hospital contributes a local likelihood factor $p(\mathcal{D}_m \mid \theta_m)$, and the global learning objective aggregates these local contributions through optimization rather than explicit Bayesian updating.

Accordingly, we can reinterpret the following regularized version of (1),

$$\arg \min_{\theta_m, \theta_g} \sum_{m=1}^M l(\theta_m; \mathcal{D}_m) + \alpha \|\theta_m - \theta_g\|, \quad (3)$$

as a Bayesian (ELBO-like) objective:

$$\arg \min_{\theta_m, q_\theta(\theta_m)} KL(q_\theta(\theta_m) \parallel p(\theta_m)) - \mathbb{E}_{q_\theta(\theta_m)}[\log p(\mathcal{D}_m \mid \theta_m)], \quad (4)$$

where the KL term corresponds to a prior-based regularizer and the expected log-likelihood reflects the empirical loss. This formulation provides a Bayesian interpretation of the underlying optimization problem.

Privacy Vulnerabilities. Despite the apparent privacy afforded by federated learning, where different parties jointly collaborate without giving up their data, a growing body of work has identified concrete privacy vulnerabilities in naïve FL. For example, gradient and model updates shared during training can be exploited to reconstruct private training data via gradient inversion attacks, even when raw data are never transmitted explicitly [Geiping et al., 2020]. Similarly, membership inference attacks allow adversaries to determine whether specific data points were used in a client’s training set [Melis et al., 2019]. Surveys of the FL threat landscape also catalogue a variety of attack classes—including data reconstruction, model inversion, and more subtle inference attacks—that undermine the premise that FL inherently protects participant data [Lyu et al., 2020].

These vulnerabilities arise because shared updates implicitly carry information about local data distributions, and because standard FL protocols lack strong cryptographic protection or rigorous privacy guarantees. Defense strategies such as differential privacy introduce trade-offs in utility and communication and often rely on loose or data-dependent bounds. Consequently, improving the theoretical privacy guarantees of these defenses remains an open challenge. Before discussing recent improvements, we must describe practical techniques for Bayesian federated learning.

1.2 Markov Chain Monte Carlo

The Bayesian reinterpretation of our federated objective contains terms like $\mathbb{E}_{q_\theta(\theta_m)}[\log p(\mathcal{D}_m \mid \theta_m)]$ which are rarely available in closed form, especially when θ_m parameterizes a complex model such as a neural network. A common way to approximate such quantities is through Monte Carlo estimation, which in turn requires the ability to draw samples from distributions involving $p(\mathcal{D}_m \mid \theta_m)$.

Markov Chain Monte Carlo (MCMC) methods provide a general method for obtaining such samples by constructing a Markov chain whose stationary distribution matches the target density. Although originally developed for exact Bayesian posterior inference, here MCMC serves as a practical tool for sampling from the distributions that appear in our ELBO-like objective and for approximating otherwise intractable expectations.

MCMC methods allow sampling from a target distribution $\mu(\theta)$ known only up to a normalizing constant, such as the posterior

$$p(\theta \mid \mathcal{D}) \propto p(\theta) p(\mathcal{D} \mid \theta).$$

Rather than attempting to evaluate the evidence Z explicitly, MCMC constructs a Markov chain $\{\theta^{(k)}\}_{k \geq 0}$ on the parameter space $\Theta \subseteq \mathbb{R}^d$ whose transition kernel $K(\theta, \theta')$ admits μ as its stationary distribution; that is,

$$\mu(\theta') = \int_{\Theta} \mu(\theta) K(\theta, \theta') d\theta \quad \text{for all } \theta' \in \Theta.$$

Under mild conditions such as irreducibility, aperiodicity, and positive recurrence, the chain is ergodic and satisfies the convergence

$$\theta^{(k)} \xrightarrow{\text{dist.}} \mu, \quad \frac{1}{K} \sum_{k=1}^K f(\theta^{(k)}) \xrightarrow{a.s.} \mathbb{E}_{\mu}[f(\theta)] \quad \text{for bounded measurable } f.$$

A common mechanism for ensuring μ is stationary for the chain is the detailed balance (reversibility) condition,

$$\mu(\theta) K(\theta, \theta') = \mu(\theta') K(\theta', \theta) \quad \text{for all } \theta, \theta' \in \Theta,$$

which guarantees that μ is invariant under the action of K .

Classical MCMC algorithms implement concrete forms of such kernels. For example, the Metropolis–Hastings (MH) method begins with a proposal distribution $q(\theta' | \theta)$ and accepts the proposed state θ' with probability

$$\alpha(\theta, \theta') := \min \left\{ 1, \frac{\mu(\theta') q(\theta | \theta')}{\mu(\theta) q(\theta' | \theta)} \right\}.$$

Gibbs sampling can be viewed as a special case of MH in which the proposal samples exactly from the full conditional distribution of a coordinate or block of coordinates.

While these methods are asymptotically exact, their practical performance can degrade severely in high-dimensional or large-data regimes. In particular, each transition may require evaluating the full data likelihood $p(\mathcal{D} | \theta)$, and random-walk-type proposals can mix poorly when μ is concentrated around a complicated manifold or has strong correlations between coordinates. These limitations motivate gradient-based MCMC methods, which leverage smoothness of $U(\theta) := -\log \mu(\theta)$ to propose more informed transitions. Such methods reduce random-walk behavior and often admit continuous-time interpretations via stochastic differential equations.

1.3 Stochastic gradient Langevin dynamics and its federated variants

It is useful to approach gradient-based MCMC from its continuous-time foundations. Whereas Metropolis–Hastings relies on generic proposal mechanisms that may mix poorly in high dimensions or when the target distribution exhibits strong correlations, diffusion-based samplers such as Langevin and dynamics construct proposals by following the gradient of $U(\theta) := -\log \mu(\theta)$.

In order to sample from posterior $p(\theta | \mathcal{D}) \propto \exp(-U(\theta))$, which generalizes to any target distribution with density that is positive and differentiable on its support, we first define the stochastic differential equation, or SDE (see Appendix A.2 for more detail)

$$d\theta_t = -\frac{1}{2} \nabla U(\theta_t) dt + dW_t = \frac{1}{2} \nabla \log p(\theta_t | \mathcal{D}) dt + dW_t,$$

where W_t is Brownian motion.² Intuitively, the gradient pushes the model to high-probability regions while the noise allows it to explore.

It can be shown that the stationary distribution of the above SDE is simply the posterior $p(\theta | \mathcal{D})$:

Theorem 1.1 (Convergence of Langevin diffusion to the posterior (informal)). *Let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be a sufficiently smooth potential satisfying standard regularity and dissipativity assumptions ensuring well-posedness of the Langevin SDE*

$$d\theta_t = -\frac{1}{2} \nabla U(\theta_t) dt + dW_t.$$

Let μ be the probability density proportional to e^{-U} ,

$$\mu(x) = Z^{-1} e^{-U(x)}, \quad Z < \infty.$$

Then:

- (a) μ is the unique invariant distribution of the diffusion.
- (b) $\mathcal{L}(\theta_t) \rightarrow \mu$ as $t \rightarrow \infty$ in total variation.
- (c) If U is strongly convex (e.g. $\nabla^2 U \succeq mI_d$), the convergence is exponentially fast.

In particular, for $U(\theta) = -\log p(\theta | \mathcal{D})$, the diffusion has $p(\theta | \mathcal{D})$ as its unique stationary distribution and converges to it over time.

²Brownian motion is the limit as $n \rightarrow \infty$ of symmetric random walks W_t^n discretized to n steps in time interval t with $W_0^n = 0$ and $W_{k+1} = W_k + \sqrt{\frac{t}{n}}$ with probability $\frac{1}{2}$, $W_{k+1} = W_k - \sqrt{\frac{t}{n}}$ with probability $\frac{1}{2}$. A more thorough treatment of Brownian motion is in Appendix A.1.

A more precise theorem statement is in Appendix A.3.

Since we can't simulate the SDE exactly, we discretize as follows³:

$$\theta' = \theta + \frac{\epsilon}{2} \nabla \log p(\theta \mid \mathcal{D}) + \sqrt{\epsilon} \eta, \quad \eta \sim \mathcal{N}(0, I). \quad (5)$$

While this sampling approach converges to the posterior, evaluating the term

$$\nabla \log p(\theta \mid \mathcal{D}) = \nabla \log p(\theta) + \sum_{i=1}^n \nabla \log p(x_i \mid \theta)$$

involves summing over all datapoints, presenting computational challenges when optimizing over large datasets. To address this computational inefficiency, there is a separate class of gradient-based methods called stochastic optimization applying the update below at each iteration t given some subset of data points $X_t = \{x_{t1}, \dots, x_{tn}\}$:

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left(\nabla \log p(\theta_t) + \frac{n}{n_t} \sum_{i=1}^n \nabla \log p(x_{ti} \mid \theta_t) \right) \quad (6)$$

subject to the conditions

$$\sum_{t=1}^{\infty} \epsilon_t = \infty \quad \sum_{t=1}^{\infty} \epsilon_t^2 < \infty,$$

to ensure convergence to a maximum. The subset gradients approximate the global gradient, and the noise introduced in each step averages out over the course of optimization. However, stochastic optimization methods fail to model uncertainty and, as a result, are prone to overfitting.

Welling and Teh [2011] marry the computational efficiency of stochastic gradient updates (6) updates with the good probabilistic properties of Langevin sampling (5) via stochastic gradient Langevin dynamics. The authors combine the two techniques according to the following update:

$$\Delta\theta_t = \frac{\epsilon_t}{2} (\nabla \log p(\theta_t) + \frac{n}{n_t} \sum_{i=1}^{n_t} \nabla \log p(x_{ti} \mid \theta_t)) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \epsilon_t). \quad (7)$$

The contribution of gradient noise becomes negligible relative to the Brownian noise, and so updates (7) approach (5) as $t \rightarrow \infty$. In particular, the variance of the noise from stochastic updates scales as $(\frac{\epsilon_t}{2})^2 V(\theta_t)$, which is dominated by the Gaussian noise with variance ϵ_t for large t (here, $V(\theta_t)$ is just the finite variance of the stochastic gradient random variable). In this sense, SGLD can be viewed as a computationally efficient minibatch approximation of Langevin dynamics that retains the correct asymptotic stationary distribution while operating on large datasets.

Distributed SGLD. In the hopes of improving optimization speed by running computations in parallel, Ahn et al. [2014] adapt SGLD to the decentralized setting. They exploit a natural form of parallelism inherent to mini-batch-based MCMC methods. Because SGLD relies on subsampling, a Markov chain can be passed between clients, drawing mini-batches exclusively from each hospital independently. To obtain an unbiased estimator of the global gradient, the gradient estimate is rescaled according to the hospital size and its sampling frequency. Formally, if hospital s has n_s datapoints and is selected with probability f_s , then the corrected stochastic gradient estimator is

$$v_s(\theta) = \nabla \log p(\theta) + \frac{n_s}{f_s m} \sum_{x \in x_s^{(m)}} \nabla \log p(x \mid \theta),$$

where $x_s^{(m)}$ denotes a mini-batch of size m sampled from hospital s . This update is then inserted into the SGLD step

$$\theta_{t+1} = \theta_t + \frac{\epsilon_t}{2} v_s(\theta_t) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \epsilon_t I).$$

³Normally, this discretization would introduce bias to be remedied by a Metropolis-Hastings correction, but the improved variant we consider mitigates this bias.

Federated SGLD. While DSGLD is effective in distributed i.i.d. settings, its performance suffers in federated environments with heterogeneous local datasets and infrequent communication between clients. Non-i.i.d. hospital distributions drastically increase the variance of DSGLD’s stochastic gradient estimator, slowing mixing and causing the chain to converge to a mixture of local posteriors rather than the true global posterior.

To address this, el Mekkaoui et al. introduce conducive gradients, a lightweight correction term that incorporates a client-specific surrogate likelihood approximation $q_s(\theta)$ and its global aggregation $q(\theta) = \prod_s q_s(\theta)$. The resulting FSGLD update augments DSGLD’s estimator by

$$g_s(\theta) = \nabla \log q(\theta) - \frac{1}{f_s} \nabla \log q_s(\theta),$$

giving the modified step

$$\theta_{t+1} = \theta_t + \frac{\epsilon_t}{2} \left(v_s(\theta_t) + g_s(\theta_t) \right) + \eta_t.$$

These conducive gradients introduce negligible overhead: each client computes q_s once at the start, communicates it a single time, and thereafter applies $g_s(\theta)$ at every local update. Intuitively, the correction reduces between-client gradient mismatch by pulling local updates toward regions where the global posterior density is plausibly high. As a result, F-SGLD remains stable under severe non-IID data and long communication intervals.

Formally, el Mekkaoui et al. [2021] compare bounds on the mean-squared error (MSE) of both DSGLD and its federated variant. The following result summarizes the behavior of standard DSGLD in heterogeneous settings:

Theorem 1.2. *Let $h_t = h$ for all $t \in \{1, \dots, T\}$. Under standard regularity conditions and assuming the gradient of the log-likelihood on each hospital is uniformly bounded,*

$$\|\nabla \log p(x_i | \theta)\| \leq \gamma_s \quad \text{for all } \theta \text{ and } x_i \in \mathbf{x}_s, \text{ each } s \in \{1, \dots, S\}, \quad (8)$$

the MSE of DSGLD for a smooth test function ϕ at time $K = hT$ is bounded, for some constant C independent of T and h , by

$$\mathbb{E}(\hat{\phi} - \bar{\phi})^2 \leq C \left(\frac{\sum_s \frac{n_s^2}{f_s} \gamma_s^2}{mT} + \frac{1}{hT} + h^2 \right).$$

The bound shows that when hospital distributions differ substantially (e.g. larger γ_s or highly imbalanced n_s/f_s) the variance term can dominate, slowing convergence and preventing the chain from accurately approximating the global posterior.

This motivates the need for variance-reduction mechanisms. FSGLD achieves this through its conducive-gradient correction, and the next theorem formalizes the resulting improvement. Here, the variance term no longer depends directly on the raw gradient heterogeneity within each hospital, but instead on the mismatch between each local surrogate q_s and its contribution to the global approximation—captured by ϵ_s , the Lipschitz surrogate error for client s . When the surrogates are well chosen (e.g., smooth and reasonably calibrated to local likelihoods), this error is significantly smaller than the variability inherent in the non-i.i.d. data itself.

Theorem 1.3. *Let $h_t = h$ for all $t \in \{1, \dots, T\}$. Assume $\log q_1, \dots, \log q_s$ are Lipschitz continuous. Under standard regularity assumptions and Assumption 8, the MSE of F-SGLD for a smooth test function ϕ at time $K = hT$ is bounded, for some constant C independent of T and h , by*

$$\mathbb{E}(\hat{\phi} - \bar{\phi})^2 \leq C \left(\frac{1}{mT} \sum_s \frac{N_s^2}{f_s} \epsilon_s^2 + \frac{1}{hT} + h^2 \right).$$

Since the surrogate-model mismatch can be controlled or minimized through lightweight surrogate construction, the algorithm avoids the variance explosion that limits D-SGLD. Consequently, F-SGLD can mix and converge accurately even when clients hold sharply different data distributions and communicate infrequently.

1.4 History of private Langevin-based sampling

A central challenge in the privacy analysis of Langevin-type sampling algorithms is that stepwise DP accounting treats each stochastic update as an independent privacy mechanism, yielding privacy bounds that scale linearly with the number of iterations. Recent work challenges this perspective, revealing that the natural dynamics of the algorithm itself—its contraction, randomness, and sampling structure—can substantially amplify privacy. Below we review this line of work and explain how our analysis builds on the existing theory.

Privacy amplification from diffusion dynamics. Chourasia et al. [2021] develop the first analysis that views the privacy loss of noisy GD and Langevin diffusion as a dynamical quantity. Rather than applying composition to per-step Gaussian mechanisms, they instead compare the laws of two coupled diffusions using Girsanov’s theorem. Under strong convexity and smoothness of the loss, their algorithm yields exponentially decaying Rényi privacy loss and thus dramatically tighter bounds than those obtained via composition. However, this analysis requires full-batch gradients and strong convexity and does not apply to stochastic gradient or nonconvex regimes.

Constant-time privacy for noisy SGD. Altschuler and Talwar [2023] demonstrate that even without strong convexity, noisy SGD on a bounded domain enjoys constant-time privacy: after a short “burn-in” period on the order of n , further iterations incur no additional privacy loss. Their argument blends two amplification mechanisms: (i) privacy amplification by iteration, arising from the contractive behavior of the gradient step, and (ii) privacy amplification by sampling, due to the minibatch subsampling inherent in SGD. The analysis is geometric and relies on optimal-transport tools rather than diffusion couplings. These powerful results assume smooth convexity, bounded diameter of the state space, and sufficiently small step size.

Uniform-in-time privacy for MCMC algorithms. Most recently, Bertazzi et al. [2025] developed a Girsanov-based technique for obtaining uniform-in-time (ϵ, δ) and Rényi-DP guarantees for diffusion-driven MCMC methods, including ULA and SGLD. Their analysis compares the law of the entire diffusion path under two adjacent datasets by introducing a bounded drift perturbation on the last unit of time and applying Girsanov’s theorem to relate the likelihoods. While remarkably general and applicable even in nonconvex settings, the resulting privacy bounds are expressed in terms of worst-case pathwise quantities. In particular, their specialized SGLD result does not reflect any privacy amplification from subsampling.

Privacy amplification through hidden-state subsampling. Recent work by Ye and Shokri [2022] shows that the privacy loss of noisy SGD can be dramatically reduced when the algorithm’s internal state is hidden and the analysis properly accounts for the stochasticity of mini-batch sampling. They derive converging Rényi DP bounds by exploiting two amplification effects: (i) randomized post-processing from injected noise, and (ii) repeated subsampling of data across epochs. Their results reveal that composition severely overestimates the privacy cost of stochastic-gradient methods, especially when convergence is slow. This perspective is directly relevant to our setting: SGLD and its federated variants also perform iterative noisy updates on randomly subsampled data, yet existing diffusion-based analyses (e.g., Bertazzi et al.) ignore this amplification and treat drift discrepancies in a worst-case manner. Our goal is to incorporate the kind of hidden-state, subsampling-based amplification identified by Ye and Shokri into a Girsanov-style analysis tailored to (federated) Langevin dynamics.

1.5 Our contributions

Centralized SGLD with subsampling amplification. We first revisit minibatch SGLD in the centralized IID setting. Bertazzi et al. obtain a uniform-in-time RDP guarantee for the final iterate of full-gradient Langevin dynamics, and then extend this to SGLD using worst-case bounds on the coupled trajectories that do not reflect minibatch subsampling. In Theorem 3.2, we adapt their perturbation construction but keep track of the randomness coming from minibatch sampling. Using standard concentration inequalities for the coupling error, we show that the final-iterate mechanism of SGLD is $(\alpha, \epsilon_\alpha)$ -RDP with $\epsilon_\alpha = O(q)$ while remaining uniform in the number of iterations T and allowing a nonconvex data term with a strongly convex regularizer. This addresses an open question raised by Bertazzi et al. about how their law-level bounds depend on the minibatch fraction q .

Law-level privacy for distributed SGLD. We next move to a distributed setting in which data are partitioned across S hospitals, with IID splits and a scheduler that selects hospitals according to fixed frequencies $(f_1, \dots, f_S)$. We study a distributed SGLD (DSGLD) scheme where, at each iteration, the server chooses a hospital i and that hospital runs one local SGLD update using a minibatch of size s from its own records before sending the update back. In Theorem 4.1, we show that the final-iterate mechanism is uniformly-in-time RDP at the record level, with parameters that depend on the effective per-hospital subsampling rate

$$q_i := \frac{f_i s}{N_i}$$

in a way analogous to the centralized $q = s/m$ case. Thus our diffusion-based method captures subsampling amplification in DSGLD while still operating at the law level.

Law-level privacy for federated SGLD. Finally, we consider a federated, non-IID variant of DSGLD called FSGLD in which each hospital maintains a local surrogate of the global posterior and contributes a conducive correction term, following el Mekkaoui et al. (2021). This setting incorporates both data heterogeneity and client subsampling. Under natural smoothness and sensitivity assumptions on the local surrogates, we show in Theorem 5.2 that the FSGLD final-iterate mechanism enjoys a uniform-in-time RDP guarantee whose dependence on the scheduler and hospital sizes again enters only through the effective rates q_i . To the best of our knowledge, this is the first explicit differential privacy analysis of FSGLD.

Implications for Bayesian federated learning. Taken together, these results provide a unified Girsanov-based privacy analysis for SGLD in centralized, distributed, and federated Bayesian learning. Our bounds are uniform in time, so a single run of SGLD can be continued as long as desired without accumulating additional privacy loss beyond the limiting RDP parameter. Because the analysis is expressed directly in terms of the law of the final iterate, the same RDP bounds extend immediately to any post-processing of the sampled parameter (e.g., predictions or downstream statistics). In the distributed setting, we further observe that standard group-privacy properties of RDP [Mironov, 2017] allow us to upgrade our record-level guarantees to statements about hospital-level participation (Lemma 4.4). Finally, our results complement prior work on noisy SGD [Altschuler and Talwar, 2023, Chourasia et al., 2021] and diffusion-level privacy analyses [Ye and Shokri, 2022] by showing that SGLD-based posterior sampling in realistic federated architectures also enjoys nontrivial privacy amplification from subsampling and hidden-state noise.

2 Preliminaries

In this section, we set up the privacy framework we will work in moving forward, elaborate on the Girsanov-based method developed by Bertazzi et al. for analyzing the privacy of diffusion-based MCMC algorithms, and introduce the basic concentration tools that we will use to refine this method and adapt it to the federated SGLD setting.

2.1 Our privacy framework

We assume a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying all the randomness of our mechanisms (Brownian motions, minibatch sampling, hospital sampling, etc.), and a data domain S . A dataset of size n is a vector

$$\mathcal{D} = (d_1, \dots, d_n) \in S^n.$$

Definition 2.1 (Adjacency). Two datasets $\mathcal{D}, \mathcal{D}' \in S^n$ are adjacent if they differ in exactly one entry, i.e. there exists a unique $j \in \{1, \dots, n\}$ such that $d_j \neq d'_j$ and $d_i = d'_i$ for all $i \neq j$.

We work exclusively with this “swap-one” notion of adjacency. Let $\mathcal{X} := \bigsqcup_{n \geq 1} S^n$ be our dataset space. Then a randomized algorithm is a measurable map

$$\mathcal{A} : \mathcal{X} \times \Omega \rightarrow E,$$

where $(E, \mathcal{E})$ is some measurable output space. For each dataset $\mathcal{D}$, the output law of $\mathcal{A}$ is the pushforward measure

$$\mathbb{P}_{\mathcal{A}(\mathcal{D})}(B) := \mathbb{P}(\mathcal{A}(\mathcal{D}, \omega) \in B), \quad B \in \mathcal{E}.$$

In later sections, we take $\mathcal{A}$ to be the hidden state mechanism that releases only the final iterate of a Markov chain after some N steps, or of a stochastic differential equation after some time T .

Definition 2.2 ((ε, δ) -differential privacy). Let $\varepsilon \geq 0$ and $\delta \in [0, 1)$. A randomized algorithm $\mathcal{A}$ is (ε, δ) -differentially private if for all adjacent datasets $\mathcal{D}, \mathcal{D}'$ and all measurable $B \in \mathcal{E}$,

$$\mathbb{P}_{\mathcal{A}(\mathcal{D})}(B) \leq e^\varepsilon \mathbb{P}_{\mathcal{A}(\mathcal{D}')} (B) + \delta.$$

Although previous analyses of DP-SGD and DP-SGLD are often carried out in terms of (ε, δ) -DP by obtaining a stepwise privacy guarantee and then using strong composition (Bassily et al., 2014; Abadi et al., 2016), both our work and that of Bertazzi et al. proceed via a natural relaxation called Rényi differential privacy, converting to (ε, δ) -DP only for interpretation. We will see why that is later.

2.1.1 Radon–Nikodym derivatives

A recurring pattern in our arguments is to compare the laws of $\mathcal{A}(\mathcal{D})$ and $\mathcal{A}(\mathcal{D}')$ indirectly, by relating them to a common reference measure on $(\Omega, \mathcal{F})$.

Definition 2.3 (Absolute continuity). Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$. We say that $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ if $\mathbb{Q}(B) = 0$ for all measurable sets B for which $\mathbb{P}(B) = 0$. The short hand for this is $\mathbb{Q} \ll \mathbb{P}$.

Theorem 2.4 (Radon–Nikodym). Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$ with $\mathbb{Q} \ll \mathbb{P}$. Then there exists a nonnegative $\mathcal{F}$ -measurable random variable Z such that

$$\mathbb{Q}(A) = \int_A Z \, d\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

The random variable Z is unique up to $\mathbb{P}$ -almost sure equality and is denoted $Z = \frac{d\mathbb{Q}}{d\mathbb{P}}$.

Proof. See Appendix A.4 in Durrett (2019). □

As we will see shortly, one typically constructs, for each adjacent pair $(\mathcal{D}, \mathcal{D}')$, an auxiliary measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ such that the law of $\mathcal{A}(\mathcal{D}')$ under $\mathbb{P}$ coincides with the law of $\mathcal{A}(\mathcal{D})$ under $\mathbb{Q}$:

$$\mathbb{P}_{\mathcal{A}(\mathcal{D}')} = \mathbb{Q}_{\mathcal{A}(\mathcal{D})}.$$

The Radon–Nikodym derivative then plays the role of a likelihood ratio between the two executions, and controlling its tails is the main route to DP guarantees in the environment of Bertazzi et al.

2.1.2 Rényi divergence and Rényi differential privacy

We next recall Rényi divergence and introduce Rényi DP, which will be our primary privacy notion.

Definition 2.5 (Rényi divergence). Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$ with $\mathbb{P} \ll \mathbb{Q}$. Given $\alpha > 1$, the *Rényi divergence of order α* of $\mathbb{P}$ with respect to $\mathbb{Q}$ is

$$D_\alpha(\mathbb{P} \parallel \mathbb{Q}) := \frac{1}{\alpha - 1} \log \int_\Omega \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right)^\alpha d\mathbb{Q} = \frac{1}{\alpha - 1} \log \mathbb{E}_\mathbb{Q} \left[\left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right)^\alpha \right].$$

We can think of Rényi divergence as a measure of how distinguishable two probability measures are, since it is zero if and only if the two probability measures are almost surely equal. Indeed, for $\alpha > 1$, the moment $\mathbb{E}_\mathbb{Q}[(d\mathbb{P}/d\mathbb{Q})^\alpha]$ is dominated by the tail behavior of $d\mathbb{P}/d\mathbb{Q}$, i.e. a larger $D_\alpha(\mathbb{P} \parallel \mathbb{Q})$ means that $d\mathbb{P}/d\mathbb{Q}$ places more mass on very large values (events where $\mathbb{P}$ is much larger than $\mathbb{Q}$), and hence that $\mathbb{P}$ and $\mathbb{Q}$ are easier to distinguish statistically. Conversely, small Rényi divergence forces $d\mathbb{P}/d\mathbb{Q}$ to stay close to 1 with high probability, making the two distributions hard to tell apart.

Proposition 2.6 (Data processing inequality for Rényi divergence; van Erven and Harremoës, 2014). Let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$ with $\mathbb{P} \ll \mathbb{Q}$. For any sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$,

$$D_\alpha(\mathbb{P}|_\mathcal{G} \parallel \mathbb{Q}|_\mathcal{G}) \leq D_\alpha(\mathbb{P} \parallel \mathbb{Q}) \quad \text{for all } \alpha > 1,$$

where $\mathbb{P}|_\mathcal{G}$ denotes the probability measure $\mathbb{P}$ restricted to the space $(\Omega, \mathcal{G})$.

The data processing inequality says that discarding information cannot make two probability measures easier to distinguish. We will use it to transfer Rényi divergence bounds proved at the level of the full path measures of our diffusions to those of the output law of our final-state mechanism $\mathcal{A}$.

Definition 2.7 ((α, ε) -Rényi differential privacy). Let $\alpha > 1$ and $\varepsilon \geq 0$. A randomized algorithm $\mathcal{A}$ is (α, ε) -Rényi differentially private if for all adjacent datasets $\mathcal{D}, \mathcal{D}'$,

$$D_\alpha(\mathbb{P}_{\mathcal{A}(\mathcal{D})} \parallel \mathbb{P}_{\mathcal{A}(\mathcal{D}')}) \leq \varepsilon.$$

Theorem 2.8 (RDP implies DP; Mironov, 2017). *If $\mathcal{A}$ is (α, ε) -Rényi differentially private for some $\alpha > 1$, then for every $\delta \in (0, 1)$, $\mathcal{A}$ is also*

$$\left(\varepsilon + \frac{1}{\alpha - 1} \log \frac{1}{\delta}, \delta \right)\text{-differentially private.}$$

Because RDP implies DP, it is okay for us to work exclusively in the Rényi DP regime for this paper.

2.1.3 Differential privacy via Radon–Nikodym derivatives

We now record the RN-based DP criteria that underlie the diffusion analysis of Bertazzi et al.

Lemma 2.9 (DP via RN-derivative; Bertazzi et al., 2025). *Suppose that for every pair of adjacent datasets $\mathcal{D}, \mathcal{D}'$ there exists a measure $\mathbb{Q}$ satisfying $\mathbb{P}_{\mathcal{A}(\mathcal{D})} = \mathbb{Q}_{\mathcal{A}(\mathcal{D}')}$, and that $\frac{d\mathbb{P}}{d\mathbb{Q}}$ is well defined. If*

$$\mathbb{P}\left(\frac{d\mathbb{P}}{d\mathbb{Q}} > e^\varepsilon\right) \leq \delta$$

for some $\varepsilon \geq 0$ and $\delta \in [0, 1)$, then $\mathcal{A}$ is (ε, δ) -differentially private.

Proof. Fix adjacent datasets $\mathcal{D}, \mathcal{D}'$ and a measurable set $B \in \mathbb{E}$. Let

$$K_\varepsilon := \left\{ \omega \in \Omega : \frac{d\mathbb{P}}{d\mathbb{Q}}(\omega) > e^\varepsilon \right\}$$

be the set of ‘bad’ Radon–Nikodym values. Then

$$\begin{aligned} \mathbb{P}_{\mathcal{A}(\mathcal{D})}(B) &= \mathbb{E}_{\mathbb{P}}[\mathbf{1}_{\{\mathcal{A}(\mathcal{D}) \in B\}} \mathbf{1}_{K_\varepsilon}] + \mathbb{E}_{\mathbb{P}}[\mathbf{1}_{\{\mathcal{A}(\mathcal{D}) \in B\}} \mathbf{1}_{K_\varepsilon^c}] \\ &\leq \delta + \mathbb{E}_{\mathbb{Q}}\left[\mathbf{1}_{\{\mathcal{A}(\mathcal{D}) \in B\}} \mathbf{1}_{K_\varepsilon} \cdot \frac{d\mathbb{P}}{d\mathbb{Q}}\right] \\ &\leq \delta + e^\varepsilon \mathbb{Q}_{\mathcal{A}(\mathcal{D})}(B) \end{aligned}$$

by the fundamental bridge, $\mathbb{P}(K_\varepsilon) \leq \delta$, and the definition of the RN-derivative. $\square$

There is an analogous statement for Rényi DP, which works at the level of moments of the likelihood ratio rather than its tail.

Lemma 2.10 (Rényi DP via RN derivative; Bertazzi et al., 2025). *In the setting of Lemma 2.9, if*

$$\mathbb{E}_{\mathbb{P}}\left[\left(\frac{d\mathbb{P}}{d\mathbb{Q}}\right)^{\alpha-1}\right] \leq e^{(\alpha-1)\varepsilon}$$

for some $\alpha > 1$ and $\varepsilon \geq 0$, then $\mathcal{A}$ is (α, ε) -Rényi differentially private.

Proof. By our assumption that $\mathbb{P}_{\mathcal{A}(\mathcal{D})} = \mathbb{Q}_{\mathcal{A}(\mathcal{D}')}$ and the data processing inequality, we have that

$$D_\alpha(\mathbb{P}_{\mathcal{A}(\mathcal{D})} \parallel \mathbb{P}_{\mathcal{A}(\mathcal{D}')}) = D_\alpha(\mathbb{P}_{\mathcal{A}(\mathcal{D})} \parallel \mathbb{Q}_{\mathcal{A}(\mathcal{D})}) \leq D_\alpha(\mathbb{P} \parallel \mathbb{Q}) = \frac{1}{\alpha - 1} \log \left(\mathbb{E}_{\mathbb{P}} \left[\left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right)^{\alpha-1} \right] \right) \leq \varepsilon$$

since $\mathbb{P}_{\mathcal{A}(\mathcal{D})}$ and $\mathbb{Q}_{\mathcal{A}(\mathcal{D})}$ coincide with $\mathbb{P}_{|\mathcal{G}}$ and $\mathbb{Q}_{|\mathcal{G}}$ where $\mathcal{G} := \{\mathcal{A}(\mathcal{D}, \cdot)^{-1}(B) : B \in \mathbb{E}\}$. $\square$

The RDP condition involves an exponential moment of the likelihood ratio $\frac{d\mathbb{P}}{d\mathbb{Q}}$ rather than a tail probability of the privacy loss random variable as in the DP condition. In our Girsanov-based analysis we obtain precisely such moment bounds from control of the drift perturbation. Working with Rényi DP therefore allows us to translate these moment bounds directly into privacy parameters. For this reason, we elect to exclusively use RDP in our analysis.

2.2 Girsanov-based method for diffusion analysis

We now summarize the Girsanov-based method of Bertazzi et al. for analyzing the privacy of diffusion-based MCMC algorithms, and extract the parts we will reuse later. Throughout this subsection we fix a time horizon $T > 0$ and consider the family of SDEs

$$dX_t^{\mathcal{D}} = f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t, \quad X_0^{\mathcal{D}} = x_0 \in \mathbb{R}^d, \quad (9)$$

where

- W is a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$;
- $\kappa : [0, \infty) \rightarrow [0, \infty)$ satisfies $\kappa(t) \leq t$ and encodes time-discretization (e.g. $\kappa(t) = \gamma \lfloor t/\gamma \rfloor$ is a backwards projection onto the time grid $\{n\gamma\}_{n \in \mathbb{N}}$)
- $(\eta_t)_{t \geq 0}$ is an auxiliary process, independent of W , used to model additional randomness such as minibatch indices;
- $f_{\mathcal{D}}$ is a dataset-dependent drift. For $\kappa(t) = \gamma \lfloor t/\gamma \rfloor$, the discrete-time chain

$$x_{n+1}^{\mathcal{D}} = x_n^{\mathcal{D}} + \gamma f_{\mathcal{D}}(x_n^{\mathcal{D}}, \eta_{n\gamma}) + \sqrt{2\gamma/\beta} z_{n+1}, \quad z_{n+1} \sim \mathcal{N}(0, I_d),$$

is equal in law to $(X_{n\gamma}^{\mathcal{D}})_{n \in \mathbb{N}}$, with step size $\gamma > 0$.

For our analysis to work we need to assume the following:

Assumption 2.11. *The SDEs we consider all have unique strong solutions.*

This is reasonable to assume as Zhang (2005) has shown it to hold when κ is a projection onto a grid like $\kappa(t) = \gamma \lfloor t/\gamma \rfloor$ or $\kappa(t) = \lfloor t \rfloor$.

The mechanisms of interest in Bertazzi et al. are the *final-value mechanism*

$$\mathcal{A}_{\text{final}}(\mathcal{D}) := X_T^{\mathcal{D}},$$

and the *path mechanism* $\mathcal{A}_{\text{path}}(\mathcal{D}) := (X_t^{\mathcal{D}})_{t \in [0, T]}$ taking values in $C([0, T], \mathbb{R}^d)$. Because in federated learning, each hospital only needs to send the final iterate of their local optimization procedure to the server, and in practice there is no real benefit to sending the whole chain, we are really only interested in $\mathcal{A}_{\text{final}}$, so we will not show Bertazzi et al.'s results for $\mathcal{A}_{\text{path}}$. Worth mentioning however is that unlike in previous analysis which treat $\mathcal{A}_{\text{final}}$ as a form of post-processing on $\mathcal{A}_{\text{path}}$ that therefore shares the same DP guarantees by the well-known invariance of DP under post-processing (Bassily et al., 2014; Abadi et al., 2016), Bertazzi et al.'s analysis of $\mathcal{A}_{\text{final}}$ and $\mathcal{A}_{\text{path}}$ are for the most part orthogonal, with the privacy guarantee of the final value greater than that of $\mathcal{A}_{\text{path}}$ when $T > 1$ under stronger conditions.

2.2.1 Girsanov's theorem and bounded perturbations

The central technical tool is a version of Girsanov's theorem tailored to drift perturbations of (9).

Theorem 2.12 (Girsanov). *For two processes X^1, X^2 solving*

$$dX_t^1 = g(X_{\kappa(t)}^1, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t, \quad dX_t^2 = g(X_{\kappa(t)}^2, \eta_{\kappa(t)}) dt + u_t dt + \sqrt{2/\beta} dW_t,$$

with the same Brownian motion W and the same auxiliary process (η_t) , under a Novikov-type integrability condition on (u_t)

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \|u_t\|^2 dt \right) \right] < \infty,$$

there exists a new measure $\mathbb{Q}$ that is absolutely continuous with respect to $\mathbb{P}$ such that $(X_t^1)_{t \in [0, T]}$ under $\mathbb{Q}$ has the same law as $(X_t^2)_{t \in [0, T]}$ under $\mathbb{P}$, with its Radon-Nikodym derivative expressible as

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(\sqrt{\frac{\beta}{2}} \int_0^T \langle u_s, dW_s \rangle - \frac{\beta}{4} \int_0^T \|u_s\|^2 ds \right).$$

From Girsanov's theorem, we see that the likelihood ratio between the two diffusions is an exponential of a stochastic integral plus a quadratic term. If u_t is uniformly bounded, one can show that the stochastic integral is sub-Gaussian (see Lemma (A.8) in the Appendix), in the sense that

$$\mathbb{E}\left[\exp\left(\int_0^T \langle u_s, dW_s \rangle\right)\right] \leq \exp\left(\frac{1}{2}T \sup_{0 \leq s \leq T} \|u_s\|^2\right).$$

Combined with our RN-based criteria for RDP (Lemma 2.10), this reduces privacy analysis to constructing a bounded perturbation u_t and bounding its norm.

2.2.2 Final-value privacy via a short-time perturbation

Bertazzi et al. use Girsanov's in a slightly indirect way. For adjacent datasets $\mathcal{D}, \mathcal{D}'$, they first couple the two diffusions $(X_t^{\mathcal{D}})_{t \in [0, T]}$ and $(X_t^{\mathcal{D}'})_{t \in [0, T]}$ by running them with the same Brownian motion and minibatch process, and assume that the coupled paths satisfy

$$\sup_{0 \leq t \leq T} \|X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}\| \leq C$$

almost surely for some finite C independent of T . They also assume that the drift difference satisfies a linear growth bound

$$\|f_{\mathcal{D}}(x, \eta) - f_{\mathcal{D}'}(y, \eta)\| \leq L\|x - y\| + c$$

for all x, y, η and constants $L, c > 0$.

To compare the laws of $X_T^{\mathcal{D}}$ and $X_T^{\mathcal{D}'}$, they do not try to realize $X^{\mathcal{D}'}$ directly as a drift-perturbation of $X^{\mathcal{D}}$. Instead, they introduce an auxiliary diffusion $(Z_t)_{t \in [0, T]}$ such that

$$Z_t = X_t^{\mathcal{D}} \quad \text{for } t \leq T-1, \quad Z_T = X_T^{\mathcal{D}'}$$

almost surely. On the last unit interval $[T-1, T]$, Z evolves according to (9) with an extra drift u_t chosen so that Z_T hits $X_T^{\mathcal{D}'}$, in order for us to get that $\mathbb{P}_{Z_T} = \mathbb{P}_{X_T^{\mathcal{D}'}}$, after which it suffices to find a probability measure $\mathbb{Q}$ with $\mathbb{P}_{Z_T} = \mathbb{Q}_{X_T^{\mathcal{D}}}$. By the coupling and drift assumptions, one can ensure that

$$\sup_{T-1 \leq t \leq T} \|u_t\| \lesssim C(L+1) + c,$$

which means u_t satisfies the Novikov-type integrability condition in Theorem 2.12. Girsanov's theorem then gives us both our desired $\mathbb{Q}$ and a likelihood ratio between the laws of $(X_t^{\mathcal{D}})_{t \in [0, T]}$ and $(Z_t)_{t \in [0, T]}$ supported only on the last unit interval, and with a bounded drift perturbation. Since the length of this interval is 1, both the exponential moments and tail probabilities of the RN derivative are bounded by expressions that do not grow with T , which allows for a uniform-in-time guarantee.

Formally, this yields the following final-state privacy guarantee.

Proposition 2.13 (Privacy of the final value; Bertazzi et al., 2025). *Suppose that for all adjacent $\mathcal{D}, \mathcal{D}'$ there exist coupled solutions $(X_t^{\mathcal{D}})_{t \in [0, T]}$ and $(X_t^{\mathcal{D}'})_{t \in [0, T]}$ of (9), driven by the same Brownian motion and minibatch process, such that almost surely*

$$\sup_{0 \leq t \leq T} \|X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}\| \leq C$$

for some finite constant C independent of T . Assume also that

$$\|f_{\mathcal{D}}(x, \eta) - f_{\mathcal{D}'}(y, \eta)\| \leq L\|x - y\| + c$$

for all x, y, η and some $L, c > 0$. Then for any $T > 0$, any $\alpha > 1$, and any $\delta \in (0, 1)$, the final-value mechanism $\mathcal{A}_{\text{final}}(\mathcal{D}) = X_T^{\mathcal{D}}$ is $(\alpha, \varepsilon_{\alpha})$ -Rényi differentially private, with

$$\varepsilon_{\alpha} = \frac{\alpha}{4} C_2, \quad C_2 = \beta(C(L+1) + c)^2,$$

where C_2 is independent of T .

Proof. See Proposition 4.5 in Bertazzi et al. (2025). □

Of note is that this guarantee is uniform-in-time.

2.2.3 Application to centralized SGLD

Bertazzi et al. apply Proposition 2.13 to SGLD for Bayesian posterior sampling. They show that under some conditions the SGLD iterates admit an SDE interpolation of the form (9), and that the coupling and drift conditions in the previous subsection are satisfied.

The resulting final-iterate guarantee can be summarized as follows.

Theorem 2.14 (Centralized SGLD: privacy of the final iterate; Bertazzi et al., 2025). *Assume*

$$\ell(x, d) = v(x, d) + k(x)$$

with v, k once continuously differentiable. Moreover assume there exists $c > 0$ with $|\nabla_x v(x, d)| \leq c$ and ∇k is L -Lipschitz and a -strongly convex, i.e.

$$\|\nabla k(x) - \nabla k(y)\| \leq L\|x - y\|, \quad \langle \nabla k(x) - \nabla k(y), x - y \rangle \geq a\|x - y\|^2.$$

Consider SGLD with dataset size m , minibatch size $s \leq m$, step size $\gamma \in (0, 2a/L^2)$, and inverse temperature $\beta > 0$. Then for any $\alpha > 1$ and $\delta \in (0, 1)$, the mechanism $\mathcal{A}_{\text{final}}$ is $(\alpha, \varepsilon_\alpha)$ -RDP with

$$\varepsilon_\alpha = \frac{\alpha}{4}C_6, \quad C_6 = c^2\beta\left(\frac{2(L+1)}{a-\gamma L^2/2} + 1\right)^2$$

where C_6 is independent of the time horizon T and of the dataset size m and minibatch size s .

2.2.4 Limitations for our purposes

For our purposes, the Girsanov-based analysis above has two main limitations.

- **Worst-case coupling and drift bounds.** Both Proposition 2.13 and Theorem 2.14 are expressed in terms of global worst-case quantities: a uniform bound C on the coupling distance $\sup_{t \leq T} \|X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}\|$ and a worst-case linear growth bound $L\|x - y\| + c$ on the drift difference. These constants do not reflect how often the algorithm actually touches the differing datum because of the subsampling or how much the trajectories typically deviate.
- **No privacy amplification from subsampling.** In the SGLD application, the constant C_6 does not depend on the dataset size m or minibatch size s , so the bound does not capture any privacy amplification from minibatch subsampling. This is in contrast to the observation in Ye and Shokri (2022) that there is an $\mathcal{O}(s/m)$ subsampling amplification rate in GSLD.

In later sections we retain the Radon–Nikodym and Girsanov viewpoint, but replace these worst-case pathwise controls by bounds on averaged drift gaps, obtained via standard sub-Gaussian and Bernstein-type concentration inequalities for sums of bounded random variables. This will allow us to recover a form of privacy amplification by subsampling.

2.3 Concentration tools

Our analysis in Section 4 of the centralized SGLD mechanism requires explicit control of exponential moments of certain bounded sums and their squares. We therefore list here the various sub-Gaussian and sub-exponential results from Boucheron, Lugosi, and Massart (2013) that we will use.

Definition 2.15 (Sub-Gaussian r.v.). We say that a random variable X is *sub-Gaussian with variance proxy σ^2* and write $X \sim \text{subG}(\sigma^2)$, if $\mathbb{E}X = 0$ and its moment generating function (MGF) satisfies

$$\mathbb{E}[\exp(sX)] \leq \exp\left(\frac{\sigma^2 s^2}{2}\right) \quad \text{for all } s \in \mathbb{R}.$$

Intuitively, $X \sim \text{subG}(\sigma^2)$ means that X has tails no heavier than a centered Gaussian with variance σ^2 , at least at the level of exponential moments. The parameter σ^2 is thought of as a “variance proxy” as it controls both the size of typical fluctuations and the decay of tails.

A convenient way to obtain sub-Gaussian random variables is via boundedness.

Lemma 2.16 (Hoeffding). *If X satisfies $\mathbb{E}X = 0$ and $X \in [a, b]$ almost surely, then*

$$X \sim \text{subG}\left(\frac{(b-a)^2}{4}\right).$$

In particular, any centered bounded random variable is sub-Gaussian, with a variance proxy that depends only on the width of its support. In our setting, we will apply Lemma 2.16 to weighted sums of centered Bernoulli indicators, yielding sub-Gaussian control on the error process S_n that appears in the proof of our SGLD privacy guarantee.

We will also need to control the square of a sub-Gaussian random variable. For this it is convenient to introduce sub-exponential random variables.

Definition 2.17 (Sub-exponential r.v.). We say that a random variable X is *sub-exponential* with parameter λ^2 , and write $X \sim \text{subE}(\lambda^2)$, if $\mathbb{E}X = 0$ and its MGF satisfies

$$\mathbb{E}[\exp(sX)] \leq \exp\left(\frac{\lambda^2 s^2}{2}\right) \quad \text{for all } |s| \leq \frac{1}{\lambda}.$$

Compared to sub-Gaussian random variables, sub-exponential random variables have slightly heavier tails: their MGFs are controlled only in a neighborhood of the origin. This is still sufficient to control the exponential moments of quadratic forms of sub-Gaussian variables.

The key link we will use is the following standard fact.

Lemma 2.18. *If $X \sim \text{subG}(\sigma^2)$, then the centered square*

$$X^2 - \mathbb{E}X^2 \sim \text{subE}(16\sigma^2).$$

3 Centralized setting: privacy analysis of SGLD

Our goal is to refine the Girsanov-based analysis of Bertazzi et al. for SGLD so that it accounts for stochastic subsampling while remaining uniform in time in a non-convex setting when we release only the final iterate of SGLD. Throughout this section we work in the centralized, IID setting.

3.1 Setup and assumptions

We specialize to loss functions of the form

$$\ell(x, d) = v(x, d) + k(x), \quad x \in \mathbb{R}^d, d \in S,$$

where v is a data-dependent term and k is a regularizer. We adopt the following assumption, which slightly strengthens the one used by Bertazzi et al. while keeping the data term nonconvex.

Assumption 3.1 (Nonconvex data term + strongly convex regularizer). *There exist constants*

$$c, \Delta_v, L_v > 0$$

such that $\nabla_x v$ is uniformly bounded by c , Δ_v -sensitive in the data argument, and L_v -Lipschitz in x . Moreover, there exist constants

$$L_k, a > 0$$

such that ∇k is L_k -Lipschitz and a -strongly convex.

Under Assumption 3.1, we consider the usual minibatch SGLD update with constant stepsize $\gamma > 0$:

$$x_{n+1} = x_n - \gamma \nabla k(x_n) - \frac{\gamma}{s} \sum_{i \in A_{n+1}} \nabla_x v(x_n, d_i) + \sqrt{\frac{2\gamma}{\beta}} \xi_{n+1},$$

where $(\xi_n)_{n \geq 1}$ are i.i.d. standard Gaussian vectors, and each $A_{n+1} \subseteq \{1, \dots, m\}$ is a minibatch of size s sampled uniformly without replacement, independently across n . We write $q := s/m$ for the subsampling rate. Given a dataset $\mathcal{D}$, we denote by $(x_n^{\mathcal{D}})_{n \geq 0}$ the SGLD iterates started from a fixed deterministic initialization x_0 , and we define the final-state mechanism

$$\mathcal{A}_{\text{final}}(\mathcal{D}) := x_T^{\mathcal{D}}$$

for some fixed horizon $T \in \mathbb{N}$. As in Bertazzi et al., we embed SGLD into a continuous-time diffusion by defining a piecewise-constant interpolation $(X_t^{\mathcal{D}})_{t \geq 0}$ solving

$$dX_t^{\mathcal{D}} = f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t,$$

where $\kappa(t) := \lfloor t \rfloor$, $(\eta_t)_{t \geq 0}$ is a process with $\eta_n := A_{n+1}$, and

$$f_{\mathcal{D}}(x, \eta) := -\gamma \nabla k(x) - \frac{\gamma}{s} \sum_{i \in \eta} \nabla_x v(x, d_i).$$

By construction, $(X_n^{\mathcal{D}})_{n \in \mathbb{N}}$ has the same law as $(x_n^{\mathcal{D}})_{n \in \mathbb{N}}$ under our base probability measure $\mathbb{P}$.

3.2 Uniform-in-time RDP for SGLD

The starting point of our analysis is the perturbation construction of Bertazzi et al., which we discussed in subsection 3.2.2: given adjacent datasets $\mathcal{D}$ and $\mathcal{D}'$, one defines a perturbed diffusion $(Z_t)_{t \in [0, T]}$ driven by the same Brownian motion and minibatch process as $X^\mathcal{D}$, but with an additional drift term u_t supported on the last unit interval $[T-1, T]$. This construction satisfies

$$Z_t = X_t^\mathcal{D} \text{ for } t \leq T-1, \quad Z_T = X_T^{\mathcal{D}'} \quad \text{a.s.}$$

so that under a suitable change of measure, $X^\mathcal{D}$ and Z can be compared via the Radon–Nikodym derivative associated with u_t , which then allows us to relate the laws of $X_T^\mathcal{D}$ and $X_T^{\mathcal{D}'}$.

Lemma B.1.1 in the appendix shows that, in this setting, controlling the exponential moments of

$$\int_{T-1}^T \|u_t\|^2 dt$$

uniformly in T is enough to obtain uniform-in-time Rényi DP guarantees for the final-state mechanism $\mathcal{A}_{\text{final}}$. The main work in our proof is therefore to derive a sharp moment bound for this quantity that reflects the randomness of minibatch sampling.

Our main centralized result is the following.

Theorem 3.2 (Uniform-in-time RDP for SGLD). *Suppose Assumption 3.1 holds. Consider SGLD with minibatch size s , dataset size m , subsampling rate $q := s/m$, and stepsize $\gamma \in (0, 2a/L_k^2)$. Then for any $\alpha > 1$ satisfying*

$$\frac{\beta}{4} \alpha(\alpha - 1) \leq \frac{q(1 - \rho^2)\gamma^2 \Delta_v^2}{34A^2 s^2}, \quad (10)$$

the final-state mechanism $\mathcal{A}_{\text{final}}$ after $T > 1$ steps is $(\alpha, \varepsilon_\alpha)$ -RDP with

$$\varepsilon_\alpha \leq \beta \alpha A^2 \frac{\gamma^2 \Delta_v^2}{s^2} \left(\frac{q}{1 - \rho^2} + \frac{q^2}{(1 - \rho)^2} \right) + \frac{1}{\alpha - 1} \log \left[(1 - q) + q \exp \left(2\beta \alpha(\alpha - 1) \frac{\gamma^2 \Delta_v^2}{s^2} \right) \right],$$

where

$$A := 1 + \gamma(L_k + L_v), \quad \rho := 1 - 2\gamma a + \gamma^2 L_k^2 \in (0, 1).$$

The bound above does not depend on T , so the privacy guarantee is uniform in the number of SGLD iterations. The proof is given in the Appendix.

Several features of Theorem 3.2 are worth highlighting.

- **Uniform in time.** As in the full-gradient analysis of Bertazzi et al., the privacy loss of releasing only the final SGLD iterate does not grow with T : all dependence on T is absorbed into the diffusion coupling and the last-step perturbation.
- **Nonconvex data term.** We allow $v(\cdot, d)$ to be nonconvex in x for each d , provided the regularizer k is strongly convex and the gradients satisfy the Lipschitz and sensitivity bounds in Assumption 3.1. This matches the nonconvex regime of Bertazzi et al. up to the additional Lipschitz and sensitivity requirements on $\nabla_x v$.
- **Amplification by subsampling.** For small q , the first term in ε_α scales linearly with q and the logarithmic term behaves like $O(q)$ as well, so overall $\varepsilon_\alpha = O(q)$ as $q \rightarrow 0$ (with all other parameters fixed). This reflects privacy amplification by subsampling, which is absent from the worst-case bounds in Bertazzi et al.’s SGLD analysis.
- **Admissible range of the Rényi order.** Since $\alpha(\alpha - 1)$ grows quadratically in α , the inequality in (10) effectively upper-bounds the Rényi order by some $\alpha_{\max}$ that depends on the algorithmic and problem parameters. In RDP-based accounting, it is customary to treat α as a tunable analysis parameter and optimize the resulting $(\varepsilon(\alpha), \delta)$ tradeoff over $\alpha > 1$. Our result allows this optimization over the interval $(1, \alpha_{\max}]$. For typical choices of (γ, β) and moderately small subsampling rates q , the admissible interval already contains the range of α values that are used in practice. Larger values of α would correspond to finer control of the privacy loss tail, but they are most useful only in regimes with extremely small δ and very large noise; in those cases one can also enlarge $\alpha_{\max}$ by increasing the noise level $1/\beta$ or decreasing the stepsize γ .

Combining Theorem 3.2 with Theorem 2.8 yields uniform-in-time (ε, δ) -DP guarantees for SGLD as well, at the cost of an additional $\frac{1}{\alpha-1} \log(1/\delta)$ term.

3.3 Proof sketch and comparison with Bertazzi et al.

We outline the main ideas behind Theorem 3.2 and how they modify the approach of Bertazzi et al.

Step 1: Girsanov-based reduction to a moment bound. For adjacent datasets $\mathcal{D}, \mathcal{D}'$ we construct a perturbation u_t on $[T-1, T]$ such that $Z_t = X_t^{\mathcal{D}}$ for $t \leq T-1$ and $Z_T = X_T^{\mathcal{D}'}$ almost surely, using the same Brownian motion and minibatch process as in Bertazzi et al. The Girsanov theorem then gives a change of measure $\mathbb{Q}$ under which $X^{\mathcal{D}}$ has the same law as Z , with RN derivative

$$\frac{d\mathbb{P}}{d\mathbb{Q}} = \exp \left(-\sqrt{\frac{\beta}{2}} \int_0^T \langle u_s, dW_s \rangle - \frac{\beta}{4} \int_0^T \|u_s\|^2 ds \right).$$

As we have mentioned, Lemma 2.10 and Lemma B.1.1 show that control of the exponential moment

$$\mathbb{E}_{\mathbb{P}} \left[\exp \left(\theta \int_{T-1}^T \|u_t\|^2 dt \right) \right]$$

for θ in a neighborhood of 0 implies $(\alpha, \varepsilon_\alpha)$ -RDP with ε_α a simple function of this moment bound.

Step 2: Bounding the drift perturbation via a contraction recursion. On $[T-1, T]$, our perturbation u_t is constant, and Assumption 3.1 yields the bound

$$\|u_t\| \leq A \|e_{T-1}\| + b \mathbf{1}_{\{j \in A_T\}}, \quad b := \frac{\gamma \Delta_v}{s},$$

where $e_n := X_n^{\mathcal{D}} - X_n^{\mathcal{D}'}$ and j is the index of the differing data point. The key recursion for the discrete-time error is

$$\|e_{n+1}\| \leq \rho \|e_n\| + b \mathbf{1}_{\{j \in A_{n+1}\}},$$

coming from the strong convexity of k (which contracts differences between iterates) and the Δ_v -sensitivity of the data term (which injects discrepancy only when the differing point is in the minibatch). Unrolling this recursion shows that $\|e_n\|$ is dominated by a weighted sum of independent Bernoulli variables indicating whether the sensitive point is sampled at each step.

Step 3: Concentration for the coupling error. At this point our analysis non-trivially departs from Bertazzi et al. They implicitly upper bound the Bernoulli indicators by 1 and use a deterministic geometric-series bound on $\|e_n\|$, which erases any dependence on the subsampling rate q . Instead, we retain the randomness of these indicators and apply the sub-Gaussian and sub-exponential tools from subsection 3.3 to obtain an MGF bound of the form

$$\mathbb{E}[\exp(\lambda \|e_{T-1}\|^2)] \leq \exp(L_1 \lambda + L_2 \lambda^2)$$

for $|\lambda|$ in a range determined by q, γ , and the problem constants because the MGF of a sub-exponential is only bounded within a specific neighborhood of 0. By restricting to λ small enough so that the quadratic term is dominated by the linear term, we obtain an essentially linear dependence on q in the exponent. Combining the bound on $\|e_{T-1}\|^2$ with the simple Bernoulli contribution from $\mathbf{1}_{\{j \in A_T\}}$ yields the MGF condition required by Lemma B.1.1, with an explicit function $\Psi(\theta)$ and an admissible range of θ that translates into the constraint on α in Theorem 3.2. Plugging this into Lemma B.1.1 then gives the claimed RDP parameters.

Comparison with prior work. From a high level, our proof follows the same Girsanov-and-perturbation template as Bertazzi et al.: we compare coupled diffusions via a Radon–Nikodym derivative and express the privacy loss in terms of a drift perturbation on the final unit interval. The difference lies in how we control this perturbation. Rather than using a worst-case bound on the coupling distance $\sup_t \|X_t^{\mathcal{D}} - X_t^{\mathcal{D}'}\|$, we exploit the specific structure of SGLD:

- the strongly convex regularizer k contracts the distance between iterates regardless of data;
- the nonconvex data term v only creates discrepancy when the sensitive point is sampled, with probability q at each step.

This combination leads to a concentration-based control of the coupling error and reveals the expected $O(q)$ privacy amplification by subsampling within our Radon–Nikodym. In contrast, the SGLD bounds in Bertazzi et al. remain insensitive to q , precisely because the subsampling randomness is not fully exploited in their coupling argument.

3.4 A Bernstein refinement

The derivation of Theorem 3.2 uses only the boundedness-based Hoeffding lemma to control

$$S_n := \sum_{k=1}^n c_k (\mathbf{1}_{\{j \in A_k\}} - q), \quad c_k := \rho^{n-k},$$

and hence obtains a sub-Gaussian parameter that does not depend on the variance $q(1-q)$ of the Bernoulli indicators. As a result, the admissible range of α in Theorem 3.2 is somewhat conservative.

A slightly sharper analysis, which we do in the Appendix, replaces Hoeffding's lemma by a Bernstein-type MGF bound for sums of bounded random variables, which tracks both the variance $q(1-q)$ and the almost-sure bound on each $c_k(\mathbf{1}_{\{j \in A_k\}} - q)$. This yields a sub-Gaussian proxy for S_n of the form

$$S_n \sim \text{subG} \left(\frac{C_S q(1-q)}{1-\rho^2} \right)$$

for an explicit numerical constant C_S , instead of the Hoeffding-based proxy $\text{subG}(C'_S/(1-\rho^2))$. Consequently, the quadratic coefficient L_2 in the MGF bound

$$\mathbb{E}[\exp(\lambda \|e_{T-1}\|^2)] \leq \exp(L_1 \lambda + L_2 \lambda^2)$$

can be reduced by a factor of order $q(1-q)$, which enlarges the interval of λ (and hence of θ and α) for which our Girsanov-based argument applies.

The overall effect is a refinement of Theorem 3.2 in which:

- the functional form of ε_α and its linear $O(q)$ scaling remain the same, but
- the admissible range of α is expanded, with the upper bound on $\frac{\beta}{4}\alpha(\alpha-1)$ improved by an explicit factor depending on $q(1-q)$.

In the next section we will see that the same ideas extend, with additional bookkeeping, to distributed and federated variants of SGLD where data are partitioned across clients.

4 IID distributed setting: privacy analysis of DSGLD

We now extend our Girsanov-based analysis to the IID distributed setting, where the dataset is partitioned across multiple hospitals and the chain is updated using a hospital scheduler. This setting should bridge between centralised SGLD and the fully federated FSGLD setting, whereby the law of the data is unchanged, but the sampling geometry is modified a bit. Our goal is to quantify how this two-level subsampling (over hospitals and over local minibatches) amplifies privacy while retaining a uniform-in-time guarantee for the final iterate. We focus on the DSGLD algorithm of Ahn et al. because its update can be written exactly as a central SGLD step with a randomly reweighted gradient estimator, so that the same diffusion interpolation and Girsanov method applies naturally.

4.1 DSGLD setup

Continue to work under Assumption 3.1. Saying $\mathcal{D}$ is partitioned into S disjoint hospitals $\mathcal{D}^{(i)}$ with size $N_i := |\mathcal{D}^{(i)}|$ where $m := \sum_{i=1}^S N_i$, at iteration $n+1$ we perform two levels of randomisation:

1. We draw a hospital index $S_{n+1} \in \{1, \dots, S\}$ according to a fixed scheduler $(f_1, \dots, f_S)$, i.e. $\mathbb{P}(S_{n+1} = i) = f_i$ with $\sum_i f_i = 1$.
2. Conditionally on $S_{n+1} = i$, we sample a minibatch $B_{n+1} \subset \mathcal{D}^{(i)}$ of size s uniformly without replacement.

The distributed SGLD (DSGLD) update takes the form

$$x_{n+1} = x_n - \gamma \nabla k(x_n) - \gamma \frac{N_{S_{n+1}}}{f_{S_{n+1}} s} \sum_{d \in B_{n+1}} \nabla_x v(x_n, d) + \sqrt{\frac{2\gamma}{\beta}} z_{n+1}, \quad z_{n+1} \sim \mathcal{N}(0, I_d), \quad (11)$$

where scaling factor $\frac{N_i}{f_i s}$ ensures the stochastic gradient estimator is unbiased for the global gradient.

We again consider a fixed horizon $T \in \mathbb{N}$ and define the final-state mechanism

$$\mathcal{A}_{\text{final}}(\mathcal{D}) := x_T^{\mathcal{D}},$$

where $(x_n^{\mathcal{D}})_{n \geq 0}$ denotes the DSGLD iterates started from a fixed $x_0 \in \mathbb{R}^d$.

4.2 Extending uniform-in-time RDP to DSGLD

We now say $\mathcal{D}, \mathcal{D}'$ are adjacent, differing at a record belonging to a unique hospital i , and we denote the changed index within $\mathcal{D}^{(i)}$ by j . At each iteration, this record is used if and only if both $S_{n+1} = i$ and $j \in B_{n+1}$. The corresponding per-iteration inclusion probability is

$$q_i := \mathbb{P}(j \in B_{n+1}, S_{n+1} = i) = f_i \frac{s}{N_i},$$

which captures the effective subsampling rate for records hosted at hospital i .

As in Section 3, we embed DSGLD into a continuous-time SDE interpolation driven by a Brownian motion and an auxiliary process encoding the scheduler and local minibatches, and then study the privacy of the final value via a Girsanov change of measure. Our distributed result mirrors Theorem 3.2, but with hospital-specific subsampling rates and gradient scales.

Theorem 4.1 (Uniform-in-time RDP for distributed SGLD). *Suppose Assumption 3.1 holds and consider DSGLD with update (11), partition $\mathcal{D} = \bigsqcup_{i=1}^S \mathcal{D}^{(i)}$, minibatch size s , scheduler $(f_1, \dots, f_S)$, and step size $\gamma \in (0, 2\alpha/L_k^2)$. We again take $A := 1 + \gamma(L_k + L_v)$, $\rho := 1 - 2\gamma a + \gamma^2 L_k^2 \in (0, 1)$ and for each hospital $i \in \{1, \dots, S\}$ we define*

$$b_i := \gamma \frac{f_i}{sN_i} \Delta_v, \quad q_i := f_i \frac{s}{N_i},$$

Then for any $\alpha > 1$ satisfying

$$\frac{\beta}{4} \alpha(\alpha - 1) \leq \min_{1 \leq i \leq S} \frac{q_i(1 - \rho^2)b_i^2}{34A^2}, \quad (12)$$

the final-state mechanism $\mathcal{A}_{\text{final}}$ after $T > 1$ iterations is $(\alpha, \varepsilon_\alpha)$ -RDP with

$$\varepsilon_\alpha \leq \max_{1 \leq i \leq S} \varepsilon_\alpha(i),$$

where, for each hospital i ,

$$\varepsilon_\alpha(i) \leq \beta \alpha A^2 b_i^2 \left(\frac{q_i}{1 - \rho^2} + \frac{q_i^2}{(1 - \rho)^2} \right) + \frac{1}{\alpha - 1} \log \left[(1 - q_i) + q_i \exp(2\beta \alpha(\alpha - 1)b_i^2) \right]. \quad (13)$$

The bound ε_α is independent of the time horizon T , so releasing only the final iterate incurs a privacy loss that is uniform in the number of DSGLD iterations. The proof is given in the Appendix.

We can note that Theorem 4.1 exhibits the following:

- **Uniform in time.**
- **Two-level amplification.** For each hospital i , the RDP parameter $\varepsilon_\alpha(i)$ exhibits the same $O(q_i)$ scaling in the effective inclusion probability q_i as in the centralized setting, but here q_i already incorporates both hospital-level scheduling (f_i) and local minibatching (s/N_i). Thus the analysis captures amplification from both levels of subsampling. Notably in the IID regime, each individual record in hospital i is included with probability q_i and the scheduler cannot distinguish whether a given inclusion arises “because of” the hospital-level or minibatch-level randomisation. It is therefore natural to ask whether the hierarchy induces any amplification beyond what is already captured by q_i ; our bound shows that for DSGLD the RDP profile depends on the hierarchy only through the pair (q_i, b_i) , with no hidden gain beyond the effective inclusion probability and gradient scale.
- **Worst-case hospital dependence.** The global RDP guarantee is driven by the worst hospital in terms of the pair (q_i, b_i) . In particular, hospitals with rare scheduling (f_i small) or large local datasets (N_i large) enjoy smaller q_i but larger b_i , and their overall contribution is governed by the combination appearing in (12) and (13).

- **Compatibility with the centralized bound.** When $S = 1$, $f_1 = 1$, and $N_1 = m$, we have $q_1 = q$, $b_1 = \gamma(m/s)\Delta_v$, and $\varepsilon_\alpha(1)$ reduces to the centralized expression in Theorem 3.2. In this sense, Theorem 4.1 strictly extends the centralized result to a distributed environment.

Combining Theorem 4.1 with Theorem 2.8 once again yields uniform-in-time (ε, δ) -DP guarantees for DSGLD by optimising over α in the admissible range (12).

4.3 Proof sketch and some modifications

The proof of Theorem 4.1 closely follows that of Theorem 3.2, with two key modifications to account for the distributed data partition and hospital scheduler. We briefly outline these.

Modification 1: Drift and interpolation with a hospital scheduler. In the distributed IID setting, the continuous-time interpolation of DSGLD is defined by

$$dX_t^{\mathcal{D}} = f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t,$$

where $\kappa(t) = \lfloor t \rfloor$, $\iota_n \in \{1, \dots, S\}$ is the hospital index at iteration n with $\mathbb{P}(\iota_n = i) = f_i$, and $\eta_n \subseteq \{1, \dots, N_{S_n}\}$ is a minibatch inside the selected hospital. For a dataset $\mathcal{D} = \bigsqcup_{i=1}^S \mathcal{D}^{(i)}$, the drift takes the form

$$f_{\mathcal{D}}(x, i, B) := -\gamma \nabla k(x) - \gamma \frac{N_i}{f_i s} \sum_{j \in B} \nabla_x v(x, d_j),$$

so that the discrete-time chain $(X_n^{\mathcal{D}})_{n \in \mathcal{N}}$ has the same law as DSGLD. The perturbation construction is unchanged: given adjacent $\mathcal{D}, \mathcal{D}'$ differing at a single point in hospital i , we define a perturbed process $(Z_t)_{t \in [0, T]}$ with drift $f_{\mathcal{D}}$ and an additional u_t supported on $[T-1, T]$ so that $Z_t = X_t^{\mathcal{D}}$ for $t \leq T-1$ and $Z_T = X_T^{\mathcal{D}'}$ almost surely. Girsanov's theorem then yields, for each i , a reference measure $\mathbb{Q}_i$ and a likelihood ratio expressed in terms of $\int_{T-1}^T \|u_t^{(i)}\|^2 dt$.

Modification 2: Error recursion and subsampling rates per hospital. Let $e_n^{(i)} := X_n^{\mathcal{D}} - X_n^{\mathcal{D}'}$ for a pair of adjacent datasets differing only in hospital i . As in the centralized proof, strong convexity and Lipschitzness of ∇k give the same contraction

$$\|T(x) - T(y)\|^2 \leq \rho \|x - y\|^2, \quad T(x) := x - \gamma \nabla k(x), \quad \rho := 1 - 2\gamma a + \gamma^2 L_k^2 \in (0, 1).$$

The only place where the hospital structure appears is in the data-dependent increment: using the Δ_v -sensitivity of $\nabla_x v$ and the reweighting $N_i/(f_i s)$ in the DSGLD drift, we obtain

$$\|e_{n+1}^{(i)}\| \leq \rho \|e_n^{(i)}\| + b_i B_{n+1}^{(i)}, \quad b_i := \gamma \frac{N_i}{f_i s} \Delta_v,$$

where

$$B_{n+1}^{(i)} := \mathbf{1}_{\{S_{n+1}=i\}} \mathbf{1}_{\{j \in B_{n+1}\}} \sim \text{Bern}(q_i), \quad q_i := \frac{f_i s}{N_i}.$$

Unrolling the recursion and comparing to an auxiliary process $\tilde{e}_n^{(i)}$ yields a weighted Bernoulli sum

$$\tilde{e}_{T-1}^{(i)} = b_i \sum_{k=1}^{T-1} \rho^{T-1-k} B_k^{(i)},$$

so that the same sub-Gaussian / sub-exponential analysis as in the centralized case applies, with (b, q) replaced by (b_i, q_i) and all constants indexed by i . The drift perturbation on $[T-1, T]$ satisfies

$$\|u_t^{(i)}\| \leq A \|e_{T-1}^{(i)}\| + b_i B_{T-1}^{(i)}, \quad A := 1 + \gamma(L_k + L_v),$$

and its exponential moments are bounded by $\Psi_i(\theta)$ depending only on (b_i, q_i) and parameters.

Plugging the bound $\mathbb{E}_{\mathbb{P}}[\exp(\theta \int_{T-1}^T \|u_t^{(i)}\|^2 dt)] \leq \exp(\Psi_i(\theta))$ into Lemma B.1 with $\theta = \frac{\beta}{4}\alpha(\alpha-1)$ yields an $(\alpha, \varepsilon_\alpha(i))$ -RDP guarantee for the final iterate under adjacency in hospital i . Taking the maximum over $i = 1, \dots, S$ gives Theorem 4.1. Apart from the reweighting by $N_i/(f_i s)$ and the

per-hospital subsampling rates q_i , the rest of the argument proceeds exactly as in the centralized setting of Theorem 3.2.

Conceptually, the DSGLD analysis shows that the uniform-in-time diffusion-level view continues to apply in distributed environments, and that the same Girsanov-based machinery cleanly captures the additional amplification induced by hospital scheduling on top of local minibatch subsampling.

From record-level to hospital-level privacy. So far, all our results in the distributed setting measure privacy at the level of individual records: datasets are adjacent when they differ in one patient. In many medical or federated applications, however, an equally natural threat model is hospital-level participation: we may wish to protect whether an entire institution joins the collaboration.

In our notation, this corresponds to an adjacency relation where $\mathcal{D} = \bigsqcup_{i=1}^S \mathcal{D}^{(i)}$ and $\mathcal{D}'$ differ only in the s th hospital, i.e. $\mathcal{D}^{(s)} \neq \mathcal{D}'^{(s)}$ while all other $\mathcal{D}^{(i)}$ coincide.

Because our core bounds are proved in terms of Rényi divergence, we can leverage the group privacy properties of RDP to upgrade record-level guarantees to hospital-level guarantees essentially for free.

Definition 4.2 (c -stable map; [Mironov, 2017]). Let $(\mathcal{D}, \sim)$ and $(\mathcal{D}', \approx)$ be two adjacency relations on datasets. A map $g : \mathcal{D}' \rightarrow \mathcal{D}$ is called c -stable if whenever $g(A)$ and $g(B)$ are adjacent in $\mathcal{D}$, there exists a sequence $D_0, \dots, D_c \in \mathcal{D}'$ such that

$$D_0 = A, \quad D_c = B, \quad \text{and} \quad D_{r-1} \approx D_r \text{ for all } r = 1, \dots, c.$$

Lemma 4.3 (Group privacy for RDP via c -stable maps; [Mironov, 2017]). Let $f : \mathcal{D} \rightarrow E$ be a mechanism that is (α, ε) -RDP with respect to an adjacency relation on $\mathcal{D}$, for some $\alpha > 1$. Let $g : \mathcal{D}' \rightarrow \mathcal{D}$ be 2^c -stable in the sense of Definition 4.2, for an integer $c \geq 1$. If $\alpha \geq 2^{c+1}$, then the composed mechanism $f \circ g : \mathcal{D}' \rightarrow E$ is $(\alpha/2^c, 3^c \varepsilon)$ -RDP. In particular, if f satisfies an RDP curve $(\alpha, \varepsilon_\alpha)$ for all $\alpha > 1$, then for any $\bar{\alpha} > 1$ and any $\alpha := 2^c \bar{\alpha} \geq 2^{c+1}$, $f \circ g$ is $(\bar{\alpha}, 3^c \varepsilon_\alpha)$ -RDP.

In our setting, changing a single hospital s modifies at most $N_s := |\mathcal{D}^{(s)}|$ records, so $c = N_s$. Applying Mironov’s result to the DSGLD mechanism gives the following hospital-level guarantee.

Lemma 4.4 (Hospital-level RDP for DSGLD). Fix a hospital $s \in \{1, \dots, S\}$ with N_s records. Suppose that for each order $\alpha > 1$ it satisfies

$$D_\alpha(\mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D})} \parallel \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')}) \leq \varepsilon_\alpha$$

whenever $\mathcal{D}, \mathcal{D}'$ differ in a single record. Let $\tilde{\mathcal{A}}_{\text{final}}$ be the same algorithm, now viewed under hospital-level adjacency, where $\mathcal{D}, \mathcal{D}'$ are adjacent if they differ only in the s th hospital. Then for any order $\bar{\alpha} > 1$ satisfying

$$\alpha := 2^{N_s} \bar{\alpha} \geq 2^{N_s+1},$$

the mechanism $\tilde{\mathcal{A}}_{\text{final}}$ is $(\bar{\alpha}, \tilde{\varepsilon}_\alpha^{(s)})$ -RDP with

$$\tilde{\varepsilon}_\alpha^{(s)} \leq 3^{N_s} \varepsilon_\alpha.$$

Proof sketch. Consider the map

$$g : (\text{datasets with hospital-level adjacency}) \longrightarrow (\text{datasets with record-level adjacency})$$

that simply reveals the underlying record-level dataset. Changing the entire s th hospital affects at most $N_s := |\mathcal{D}^{(s)}|$ records. [Mironov, 2017] shows that a map which can change at most c records at once is 2^c -stable with respect to the record-level adjacency, so in our case g is 2^{N_s} -stable. By assumption, the record-level mechanism $\mathcal{A}_{\text{final}}$ is $(\alpha, \varepsilon_\alpha)$ -RDP for every order $\alpha > 1$. Applying Lemma 4.3 with $f = \mathcal{A}_{\text{final}}$, g as above, and $c = N_s$ shows that, for any $\bar{\alpha} > 1$ such that $\alpha := 2^{N_s} \bar{\alpha} \geq 2^{N_s+1}$, the composed mechanism $\tilde{\mathcal{A}}_{\text{final}} = \mathcal{A}_{\text{final}} \circ g$ is $(\bar{\alpha}, \tilde{\varepsilon}_\alpha^{(s)})$ -RDP with $\tilde{\varepsilon}_\alpha^{(s)} \leq 3^{N_s} \varepsilon_\alpha$. $\square$

In words, Lemma 4.4 says that our DSGLD analysis controls not only the influence of any single patient, but also the influence of an entire hospital, at the cost of evaluating our record-level bound at a larger Rényi order and multiplying the privacy parameter by a factor 3^{N_s} . When hospitals hold a moderate number of records or when the primary concern is client-level participation, this provides a principled way to reason about hospital-level privacy within exactly the same framing.

5 Non-IID distributed setting: privacy analysis of federated SGLD

We now extend the DSGLD analysis of Section 4 to the federated setting, where we no longer require IID data between hospitals as is facilitated by FSGLD, and attain a qualitatively equivalent privacy scaling for FSGLD. The drift is now modified by a surrogate-based conducive gradient which is computed locally at the chosen hospital but is designed to reproduce a global surrogate gradient in expectation. This necessitates the drift Lipschitz constant and its sensitivity acquiring additional contributions from the surrogate terms, leading to slightly different constants and algebraic concessions in our RDP bounds.

5.1 FSGLD setup and assumptions

Our subsampling is equivalent to that of DSGLD.

To define FSGLD we introduce, for each hospital i and dataset $\mathcal{D}$, a hospital-wise surrogate density $\mathbf{q}_{i,\mathcal{D}}$, and set the global surrogate

$$\log \mathbf{q}_{\mathcal{D}}(x) := \sum_{r=1}^S \log \mathbf{q}_{r,\mathcal{D}}(x),$$

together with the hospital-specific conducive gradient

$$g_{i,\mathcal{D}}(x) := \nabla_x \log \mathbf{q}_{\mathcal{D}}(x) - \frac{1}{f_i} \nabla_x \log \mathbf{q}_{i,\mathcal{D}}(x).$$

Intuitively, $\mathbf{q}_{i,\mathcal{D}}$ approximates the local posterior at hospital i , while $g_{i,\mathcal{D}}$ corrects the local drift so that the hospital-averaged dynamics track the global surrogate gradient $\nabla_x \log \mathbf{q}_{\mathcal{D}}$.

We adopt the following assumption, which allows us to continue linear reductions from the same toolbox as in Bertazzi et al.

Assumption 5.1. *For each hospital $i \in \{1, \dots, S\}$ and each dataset $\mathcal{D}$, let $\{\mathbf{q}_{r,\mathcal{D}}\}_{r=1}^S$ be the hospital-wise surrogates defining the conducive gradients in FSGLD via*

$$g_{i,\mathcal{D}}(x) := \nabla_x \log \mathbf{q}_{\mathcal{D}}(x) - \frac{1}{f_i} \nabla_x \log \mathbf{q}_{i,\mathcal{D}}(x), \quad \log \mathbf{q}_{\mathcal{D}}(x) := \sum_{r=1}^S \log \mathbf{q}_{r,\mathcal{D}}(x)$$

Assume:

1. For some $L_g > 0$, uniformly in $i, \mathcal{D}$ and all $x, y \in \mathbb{R}^d$,

$$\|g_{i,\mathcal{D}}(x) - g_{i,\mathcal{D}}(y)\| \leq L_g \|x - y\|.$$

2. For each i , there exists $\Delta_i^{(q)} > 0$ such that for all adjacent datasets $\mathcal{D}, \mathcal{D}'$ differing only in the i th hospital and all $x \in \mathbb{R}^d$,

$$\|\nabla_x \log \mathbf{q}_{i,\mathcal{D}}(x) - \nabla_x \log \mathbf{q}_{i,\mathcal{D}'}(x)\| \leq \Delta_i^{(q)}.$$

From the definition of $g_{i,\mathcal{D}}$ it follows that the effective surrogate sensitivity per hospital is

$$\Delta_i^{(g)} := (f_i^{-1} - 1) \Delta_i^{(q)}.$$

The federated drift used in the continuous-time interpolation is then

$$f_{\mathcal{D}}^{\text{FS}}(x, i, B) := -\gamma \nabla k(x) - \gamma \frac{N_i}{f_i S} \sum_{j \in B} \nabla_x v(x, d_j) + \gamma g_{i,\mathcal{D}}(x), \quad (14)$$

where i denotes the hospital index and $B \subseteq x_i$ the minibatch. Compared to the distributed DSGLD drift, the extra $\gamma g_{i,\mathcal{D}}(x)$ term implements the surrogate correction. In particular, setting $g_{i,\mathcal{D}} \equiv 0$ recovers the federated DSGLD drift.

Theorem 5.2 (Uniform-in-time RDP for FSGLD). *Suppose Assumption 3.1 and Assumption 5.1 hold, and let $\gamma \in (0, 2a/L_k^2)$. We retain $\rho := 1 - 2\gamma a + \gamma^2 L_k^2 \in (0, 1)$ and define, for each hospital i ,*

$$b_i := \gamma \frac{N_i}{s f_i} \Delta_v, \quad \Delta_i^{(g)} := (f_i^{-1} - 1) \Delta_i^{(g)}, \quad c_i := \gamma \Delta_i^{(g)},$$

and the federated Lipschitz factor

$$A^{\text{FS}} := 1 + \gamma(L_k + L_v + L_g),$$

together with

$$L_1(i) := b_i^2 \left(\frac{q_i}{1 - \rho^2} + \frac{q_i^2}{(1 - \rho)^2} \right).$$

If $\alpha > 1$ satisfies

$$\frac{\beta}{4} \alpha(\alpha - 1) \leq \min_{1 \leq i \leq S} \frac{q_i(1 - \rho^2)}{51 (A^{\text{FS}})^2 b_i^2},$$

then the final-state mechanism of FSGLD after $T > 1$ steps is $(\alpha, \varepsilon_\alpha^{\text{FS}})$ -RDP with

$$\varepsilon_\alpha^{\text{FS}} := \max_{1 \leq i \leq S} \varepsilon_\alpha^{\text{FS}}(i),$$

where $\varepsilon_\alpha^{\text{FS}}(i)$ is given explicitly in Theorem 5.2. The bound has the same structure as the DSGLD bound of Theorem 4.1, with

$$A \rightsquigarrow A^{\text{FS}}, \quad \Psi_i(\theta) \rightsquigarrow \Psi_i^{\text{FS}}(\theta) := 6L_1(i) (A^{\text{FS}})^2 \theta + 3c_i^2 \theta + \log \left[(1 - q_i) + q_i e^{3\theta b_i^2} \right].$$

Proof sketch and modifications. The proof follows the coupling strategy of Theorems 3.2 and 4.1. We an adjacent pair $\mathcal{D}, \mathcal{D}'$ differing in a single datum of hospital i . We consider the continuous-time interpolations driven by $f_{\mathcal{D}}^{\text{FS}}$ and construct a perturbed process $(Z_t^i)_{t \in [0, T]}$ that coincides with $X_t^{\mathcal{D}}$ on $[0, T - 1]$ and satisfies $Z_T^i = X_T^{\mathcal{D}'}$ almost surely. As in Section 4, the RDP bound reduces to verifying the moment-generating-function condition of Lemma B.1 for the control process u_t that “patches” the step from $T - 1$ to T . The key difference is that the drift now depends explicitly on the hospital index and its surrogate, cf. (14), so both the Lipschitz constant and the sensitivity of the drift become hospital-specific quantities. Consequently we have two substantive modifications:

Modification 1: Federated drift and scheduler. On the last interval $[T - 1, T]$ we have $\kappa(t) = T - 1$ and thus

$$\|u_t\| = \|f_{\mathcal{D}'}^{\text{FS}}(X_{T-1}^{\mathcal{D}'}, S_{T-1}, B_{T-1}) - f_{\mathcal{D}}^{\text{FS}}(Z_{T-1}^i, S_{T-1}, B_{T-1}) - (Z_{T-1}^i - X_{T-1}^{\mathcal{D}'})\|.$$

We decompose the drift difference into the DSGLD part and the surrogate part:

$$\begin{aligned} & \|f_{\mathcal{D}'}^{\text{FS}}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}^{\text{FS}}(Z_{T-1}^i, \iota_{T-1}, \eta_{T-1})\| \\ & \leq \|f_{\mathcal{D}'}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}(Z_{T-1}^i, \iota_{T-1}, \eta_{T-1})\| \\ & \quad + \gamma \|g_{S_{T-1}, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'}) - g_{S_{T-1}, \mathcal{D}}(Z_{T-1}^i)\|. \end{aligned}$$

The first term is treated exactly as in the DSGLD proof, and yields

$$\|f_{\mathcal{D}'}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}(Z_{T-1}^i, \iota_{T-1}, \eta_{T-1})\| \leq \gamma(L_k + L_v) \|e_{T-1}^i\| + b_i B_{T-1}(i),$$

with $e_n^i := X_n^{\mathcal{D}} - X_n^{\mathcal{D}'}$ and $B_{T-1}(i)$ the Bernoulli indicator that the differing datum in hospital i is sampled at time $T - 1$. The only change is that the minibatch scaling is now $f_i/(sN_i)$, so b_i and q_i depend explicitly on the scheduler and hospital sizes.

Modification 2: Surrogate sensitivity and control cost. By Assumption 5.1,

$$\begin{aligned} \|g_{i, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'} - g_{i, \mathcal{D}}(Z_{T-1}^i))\| & \leq \|g_{i, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'} - g_{i, \mathcal{D}'}(Z_{T-1}^i))\| + \|g_{i, \mathcal{D}'}(Z_{T-1}^i) - g_{i, \mathcal{D}}(Z_{T-1}^i)\| \\ & \leq L_g \|e_{T-1}^i\| + \Delta_i^{(g)}. \end{aligned}$$

Combining the drift, surrogate, and linear correction terms, we obtain

$$\|u_t\| \leq (1 + \gamma(L_k + L_v + L_g)) \|e_{T-1}^i\| + b_i B_{T-1}(i) + c_i = A^{\text{FS}} \|e_{T-1}^i\| + b_i B_{T-1}(i) + c_i.$$

Squaring and using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$ introduces an extra factor 3 compared to the DSGLD case (which only involved two terms), and yields

$$\|u_t\|^2 \leq 3(A^{\text{FS}})^2 \|e_{T-1}^i\|^2 + 3b_i^2 B_{T-1}(i)^2 + 3c_i^2.$$

This is precisely where the surrogate sensitivity manifests itself in the MGF bound, via the additional deterministic term $3c_i^2$ and the replacement $A \mapsto A^{\text{FS}}$.

Then let $\theta \geq 0$ and set $\lambda := 3\theta(A^{\text{FS}})^2$. The recursion for e_n^i remains identical to DSGLD, since the surrogate correction cancels out in the difference of two coupled chains driven by the same scheduler and minibatch decisions. Hence the argument of Theorem 4.1 applies verbatim and yields

$$\mathbb{E}[\exp(\lambda \|e_{T-1}^i\|^2)] \leq \exp(2L_1(i) \lambda)$$

for $\lambda \leq q_i(1 - \rho)^2 / (17b_i^2)$. Substituting λ and using the bound on $\|u_t\|^2$, we obtain

$$\begin{aligned} \mathbb{E}\left[\exp\left(\theta \int_{T-1}^T \|u_t\|^2 dt\right)\right] &= \mathbb{E}[\exp(\theta \|u_t\|^2)] \\ &\leq \exp(6L_1(i) (A^{\text{FS}})^2 \theta) \exp(3c_i^2 \theta) [(1 - q_i) + q_i e^{3\theta b_i^2}], \end{aligned}$$

for all

$$\theta \leq \frac{q_i(1 - \rho^2)}{51(A^{\text{FS}})^2 b_i^2}.$$

Relative to DSGLD, the upper bound on θ is slightly more restrictive (the constant 51 appears instead of its non-federated counterpart) because of the additional factor 3 and the enlarged Lipschitz constant A^{FS} . This is the only place where the algebra differs: all subsequent steps in Lemma B.1 and in the derivation of the RDP bound are unchanged.

Finally, setting $\theta = \frac{\beta}{4}\alpha(\alpha - 1)$ and imposing the stated constraint on α ensures the MGF condition of Lemma B.1. Maximising over hospitals i yields the bound in Theorem 5.2, and substituting the explicit expression for $\Psi_i^{\text{FS}}(\theta)$ gives Theorem 5.2.

6 Conclusion

This work set out to understand how much privacy can be obtained from the intrinsic randomness of Langevin-type samplers in realistic federated architectures. Rather than treating stochastic gradient Langevin dynamics as a long sequence of independent mechanisms and applying composition, we adopted a law-level point of view: the algorithm is seen as a diffusion driven by injected noise and subsampling, and privacy is quantified directly in terms of how the resulting path measures change when a single record is modified. Within this framework we focused on the practically relevant mechanism that releases only the final state of the chain, which is the object actually used for prediction in federated learning.

In the centralized setting of 3, we showed that the diffusion interpolation of minibatch SGLD admits a Girsanov-type coupling between executions on adjacent datasets, and that the corresponding likelihood ratio can be controlled using simple concentration inequalities. This yields a Rényi differential privacy guarantee for the terminal iterate that does not degrade with the number of steps, even when the loss has a nonconvex data-dependent part and only the regularizer is strongly convex. Crucially, the bound depends explicitly on the subsampling rate, recovering the expected privacy amplification from minibatching within a diffusion-based analysis.

We then transported the same methodology to distributed and federated variants of SGLD. For DSGLD, where the chain hops between hospitals according to a scheduler and each update uses a local minibatch, our bounds depend on an effective inclusion probability q_i for each hospital, capturing in a single quantity the combined effect of client selection and local subsampling. In the non-IID federated regime handled by FSGLD, we further incorporated surrogate likelihoods and conducive gradients into the drift. Under mild regularity and sensitivity assumptions on these

surrogates, we obtained analogous guarantees that remain stable under heterogeneity and sparse communication. Standard group-privacy properties for Rényi DP then allow these record-level guarantees to be lifted to statements about participation of entire hospitals.

Taken together, our results demonstrate that diffusion-based privacy analyses need not ignore the structure of mini-sampled stochastic gradients. From a practical perspective, this suggests that several design choices that are already standard in Bayesian federated learning (minibatching, client sampling, and surrogate-based variance reduction) also contribute quantitatively to privacy.

6.1 Future work

This work was primarily interested in leveraging the stochastic gradient to tighten the bounds of Bertazzi et al. and to extend their method to the distributed setting. As such it remains to further tighten the Girsanov-based bound by refining the concentration inequalities applied to S_n ; in particular we do not leverage the construction of S_n as a shifted weighed sum of Bernoullis with static parameter q in creating concentration bounds (we leveraged the fact our Bernoullis were independent in our Bernstein-type bound; there is much more structure to use). We suspect there may be existing tight bounds to port from statistical theory, since S_n is a rather simple object.

Further, our construction of u_t can possibly be refined so that our variance scales with actual samplings from our differing points. We can also consider whether our extensions immediately carry to Bertazzi et al.’s work on privacy guarantees for the full path.

Despite providing a FSGLD guarantee we have not considered for what surrogate densities our guarantees are reasonable. Additionally assuming more structure on q (e.g. a toy natural model as in el Mekkaoui et al. [2021], or extending to the NEF) may allow for tighter analysis of our g term and a more favorable bound altogether.

In the distributed setting we concede the worst case for both our γ bound and our RDP guarantees; we suspect that these bounds may be contravariant and that a better bound can be realized by a less naive concession, or by looking at another form of DP that considers “averaging”/amortization. And there are of course many assumptions to relax.

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References

- Martín Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pages 308–318. ACM, 2016. doi: 10.1145/2976749.2978318.
- Sungjin Ahn, Babak Shabbaba, and Max Welling. Distributed stochastic gradient mcmc. In Eric P. Xing and Tony Jebara, editors, *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 1044–1052, Beijing, China, 22–24 Jun 2014. PMLR. URL <https://proceedings.mlr.press/v32/ahn14.html>.
- Jason M. Altschuler and Kunal Talwar. Privacy of noisy stochastic gradient descent: More iterations without more privacy loss, 2023. URL <https://arxiv.org/abs/2205.13710>.
- Borja Balle, Gilles Barthe, Marco Gaboardi, and Justin Hsu. Privacy amplification by subsampling: Tight analyses via couplings and divergences. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

- Raef Bassily, Adam Smith, and Abhradeep Thakurta. Private empirical risk minimization: Efficient algorithms and tight error bounds. In *55th IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, pages 464–473. IEEE, 2014. doi: 10.1109/FOCS.2014.56.
- Andrea Bertazzi, Tim Johnston, Gareth O. Roberts, and Alain Durmus. Differential privacy guarantees of markov chain monte carlo algorithms, 2025. URL <https://arxiv.org/abs/2502.17150>. Version cited: February 2025 preprint.
- Keith Bonawitz, Hubert Eichner, Wolfgang Grieskamp, Dzmitry Huba, Alex Ingerman, Vladimir Ivanov, Chloe Kiddon, Jakub Konečný, Stefano Mazzocchi, H. Brendan McMahan, Timon Van Overveldt, David Petrou, Daniel Ramage, and Jason Roselander. Towards federated learning at scale: System design, 2019. URL <https://arxiv.org/abs/1902.01046>.
- Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford Series in Probability and Statistics. Oxford University Press, 2013.
- Longbing Cao, Hui Chen, Xuhui Fan, Joao Gama, Yew-Soon Ong, and Vipin Kumar. Bayesian federated learning: A survey, 2023. URL <https://arxiv.org/abs/2304.13267>.
- Rishav Chourasia, Jiantao Ye, and Reza Shokri. Differential privacy dynamics of langevin diffusion and noisy gradient descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 28918–28930, 2021.
- Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5th edition, 2019.
- Khaoula el Mekkaoui, Diego Mesquita, Paul Blomstedt, and Samuel Kaski. Federated stochastic gradient langevin dynamics, 2021. URL <https://arxiv.org/abs/2004.11231>.
- Vitaly Feldman, Ilya Mironov, Kunal Talwar, and Abhradeep Thakurta. Privacy amplification by iteration. In *59th IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, pages 521–532. IEEE, 2018. doi: 10.1109/FOCS.2018.00056.
- Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. Inverting gradients – how easy is it to break privacy in federated learning?, 2020. URL <https://arxiv.org/abs/2003.14053>.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, volume 113 of *Graduate Texts in Mathematics*. Springer, second edition, 1991. doi: 10.1007/978-1-4612-0949-2. URL <https://doi.org/10.1007/978-1-4612-0949-2>.
- Tian Li, Anit Kumar Sahu, Ameet Talwalkar, and Virginia Smith. Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine*, 37(3):50–60, May 2020. ISSN 1558-0792. doi: 10.1109/msp.2020.2975749. URL <http://dx.doi.org/10.1109/MSP.2020.2975749>.
- Lingjuan Lyu, Han Yu, and Qiang Yang. Threats to federated learning: A survey, 2020. URL <https://arxiv.org/abs/2003.02133>.
- Luca Melis, Congzheng Song, Emiliano De Cristofaro, and Vitaly Shmatikov. Exploiting unintended feature leakage in collaborative learning. In *2019 IEEE Symposium on Security and Privacy (SP)*, pages 691–706, 2019. doi: 10.1109/SP.2019.00029.
- Ilya Mironov. Rényi differential privacy. In *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, pages 263–275. IEEE, 2017. doi: 10.1109/CSF.2017.11.
- Tim van Erven and Peter Harremoës. Rényi divergence and kullback–leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014. doi: 10.1109/TIT.2014.2320500.
- Max Welling and Yee Whye Teh. Bayesian learning via stochastic gradient langevin dynamics. 37(3):681–688, May 2011. ISSN 1558-0792. doi: 10.1109/msp.2020.2975749. URL <https://dl.acm.org/doi/10.5555/3104482.3104568>.

Jiantao Ye and Reza Shokri. Differentially private learning needs hidden state (or much faster convergence). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 8357–8370, 2022.

Ligeng Zhu, Zhijian Liu, and Song Han. Deep leakage from gradients, 2019. URL <https://arxiv.org/abs/1906.08935>.

A Background and Preliminaries

A.1 Brownian motion

Brownian motion describes the random movements of particles suspended in some medium. Because of its nice properties (e.g. stationary independent Gaussian increments, satisfies the Markov property, continuous sample paths, etc.), Brownian Motion serves as a natural continuous-time model of randomness in a wide range of systems, like stock price fluctuation and cellular transport.

Definition A.1 (Brownian motion). A Brownian motion is a probability measure $\mathbb{P}$ on the space of continuous functions $\Omega := C([0, \infty], \mathbb{R})$ such that the canonical process

$$W_t(\omega) := \omega(t)$$

satisfies:

1. (Initial condition) $W_0 = 0$ almost surely.
2. (Gaussian increments) For all $0 \leq s < t$, $W_t - W_s \sim \mathcal{N}(0, t - s)$.
3. (Independent increments) For any $n \geq 1$ and any sequence $0 = t_0 < t_1 < \dots < t_n$,

$$(W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}})$$

are mutually independent.

It can be shown that such a probability distribution exists, and Donsker's invariance principle motivates this distribution as a canonical choice.

Theorem A.2 (Donsker's invariance principle). *Let $X_1, X_2, X_3, \dots$ be a sequence of i.i.d random variables in $\mathbb{R}$ with $\mathbb{E}[X_i] = 0$ and $\text{Var}[X_i] = 1$. Let $S_n = X_1 + \dots + X_n$ be a partial sum of the random walk. Extend S_n to continuous time by linear interpolation:*

$$S_t = S_{[t]} + (t - [t])X_{[t]+1}, \quad t \geq 0$$

Then if we write

$$S_t^{(n)} = \frac{1}{\sqrt{n}} S_{nt}, \quad 0 \leq t \leq 1,$$

we have a convergence in law

$$S_{[0,1]}^{(n)} \xrightarrow{d} W_{[0,1]},$$

where $W_{[0,1]}$ is Brownian motion on $[0, 1]$.

Corollary A.3. $S_1^{(n)} \xrightarrow{d} W_1 \in \mathcal{N}(0, 1)$ is the Central Limit Theorem.

Donsker's invariance principle says if we take any i.i.d sequence of centered random variables each with shared constant variance, form a random walk, and rescale correctly, the resulting process converges to Brownian motion in distribution. This positions Brownian Motion as the stochastic process analogue of the Gaussian distribution.

Beyond arising as the scaling limit of random walks, Brownian motion possesses several structural properties that make it a canonical model for continuous-time randomness. In addition to having independent, stationary Gaussian increments, sample paths are almost surely continuous but nowhere differentiable with fluctuations accumulating in a precise way: its quadratic variation satisfies

$$[W]_t := \lim_{|\Pi| \rightarrow 0} \sum_k (W_{t_{k+1}} - W_{t_k})^2 = t,$$

where the limit is taken over partitions $\Pi = \{0 = t_0 < t_1 < \dots < t_m = t\}$ with mesh size $|\Pi| \rightarrow 0$. This property separates Brownian motion from smooth processes and underlies the definition of the Itô integral and Itô's lemma. Furthermore, Brownian motion is a continuous martingale, i.e. it satisfies $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ for $s \leq t$ with respect to its natural filtration $\mathcal{F}$, a feature that ensures that stochastic integrals against W_t behave analogously to sums of mean-zero random variables.

A.2 From Brownian motion to stochastic differential equations

Brownian motion provides a mathematically tractable model for continuous-time stochastic processes. Since its paths are almost surely nowhere differentiable, classical calculus techniques do not apply to expressions involving W_t . This motivates the development of Itô calculus, which provides rules for integrating with respect to Brownian motion and for analyzing stochastic differential equations.

A central object in Itô calculus is the *stochastic integral*

$$\int_0^t f(s) dW_s,$$

defined as an L^2 limit of (left) Riemann sums $\sum_k f(t_k)(W_{t_{k+1}} - W_{t_k})$. Unlike ordinary integrals, the Itô integral has mean zero and variance

$$\mathbb{E} \left[\left(\int_0^t f(s) dW_s \right)^2 \right] = \int_0^t f(s)^2 ds,$$

reflecting the quadratic variation property $[W]_t = t$ introduced earlier.

Stochastic notation such as dX_t and dW_t should not be interpreted as typical derivatives. For instance, a stochastic differential equation (SDE) of the form

$$dX_t = \mu(X_t) dt + \sigma(X_t) dW_t$$

is shorthand for the *integral equation*

$$X_t = X_0 + \int_0^t \mu(X_s) ds + \int_0^t \sigma(X_s) dW_s.$$

The first term is an ordinary (Lebesgue) integral capturing some deterministic drift, while the second term is an Itô integral representing the accumulated randomness of Brownian motion. In this sense, the symbols dt and dW_t serve as bookkeeping devices that encode the contributions of drift and noise, or diffusion, within a single differential-like notation.

This differential form (deterministic drift and Brownian diffusion) is the most straightforward combination of signal and noise. It also leads naturally to the stochastic chain rule, *Itô's lemma*: if X_t solves an SDE and g is twice differentiable, then

$$dg(X_t) = g'(X_t) dX_t + \frac{1}{2} g''(X_t) (dX_t)^2,$$

where $(dX_t)^2$ is interpreted using the quadratic variation of X_t . For example, if $dX_t = \mu_t dt + \sigma_t dW_t$, then $(dX_t)^2 = \sigma_t^2 dt$ because $(dW_t)^2 = dt$ and $dt dW_t = 0$. Heuristically, this identity can be motivated by applying a second-order Taylor expansion to g , except that, unlike in ordinary calculus, the quadratic term $\frac{1}{2} g''(X_t) (dX_t)^2$ cannot be dropped due to the non-vanishing quadratic variation of Brownian motion. For a more rigorous and comprehensive account of stochastic calculus and SDEs, see [Karatzas and Shreve, 1991].

A.3 Langevin diffusion

With Itô calculus in hand, we can define and analyze continuous-time stochastic processes of the form

$$d\theta_t = -\frac{1}{2} \nabla U(\theta_t) dt + dW_t, \tag{15}$$

known as *Langevin diffusion*. The drift term $-\frac{1}{2} \nabla U(\theta)$ pushes the process toward low values of the potential U , so that in the absence of noise, (15) would reduce to gradient flow on U ; the Brownian term injects isotropic noise to keep the process exploring the state space rather than collapsing deterministically to a single minimizer: in the long run, this causes the θ_t to spend time in each region of space in proportion to e^{-U} .

Because the Langevin update (15) is written in the differential form of an Itô SDE, one must verify that this equation indeed defines a well-posed stochastic process and that its long-term behavior corresponds to the distribution we wish to sample from. We summarize sufficient conditions for this behavior in the following theorem.

Theorem A.4 (Convergence of Langevin diffusion to the posterior). *Let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice continuously differentiable potential function, and consider the Langevin diffusion*

$$d\theta_t = -\frac{1}{2}\nabla U(\theta_t) dt + dW_t, \quad (16)$$

with initial condition $\theta_0 = \theta \in \mathbb{R}^d$. Assume:

(i) **Lipschitz drift and linear growth.** *There exists $L > 0$ such that for all $x, y \in \mathbb{R}^d$,*

$$\|\nabla U(x) - \nabla U(y)\| \leq L\|x - y\|, \quad \|\nabla U(x)\| \leq L(1 + \|x\|).$$

Then the SDE (16) admits a unique strong solution $(\theta_t)_{t \geq 0}$.

(ii) **Coercivity (dissipativity).** *There exist constants $m > 0$ and $b \geq 0$ such that for all $x \in \mathbb{R}^d$,*

$$\langle \nabla U(x), x \rangle \geq m\|x\|^2 - b.$$

Let μ be the probability density

$$\mu(x) = Z^{-1}e^{-U(x)}, \quad Z = \int_{\mathbb{R}^d} e^{-U(y)} dy < \infty.$$

Then:

(a) μ is the unique invariant (stationary) distribution of the diffusion.

(b) The distribution of θ_t converges to μ as $t \rightarrow \infty$:

$$\mathcal{L}(\theta_t) \xrightarrow[t \rightarrow \infty]{} \mu$$

in total variation (equivalently, in law).

(c) Moreover, under the stronger condition that $\nabla^2 U(x) \succeq mI_d$ for all x , the convergence is exponentially fast:

$$\|\mathcal{L}(\theta_t) - \mu\|_{\text{TV}} \leq Ce^{-mt}$$

for some constant C depending on θ_0 .

In particular, when $U(\theta) = -\log p(\theta \mid \mathcal{D})$ is the negative log-posterior, the Langevin diffusion has $p(\theta \mid \mathcal{D})$ as its unique stationary distribution and converges to it over time.

Thus, Langevin diffusion provides a sampling method for distributions of this form, including Bayesian posteriors where $U(\theta) = -\log p(\theta \mid \mathcal{D})$ up to an additive constant. This interpretation underlies the use of Langevin dynamics in Markov chain Monte Carlo methods.

A.4 Existence and uniqueness of the SDE interpolation

In Section 3.2 we introduced the family of SDEs

$$dX_t^{\mathcal{D}} = f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t, \quad X_0^{\mathcal{D}} = x_0,$$

where κ encodes a time-discretization and (η_t) is an auxiliary process. Here we justify our assumption that the SDE interpolations of SGLD and its variants admit unique strong solutions.

Assumption A.5 (Regularity of the drift). *For each dataset $\mathcal{D}$, the drift $f_{\mathcal{D}} : \mathbb{R}^d \times E \rightarrow \mathbb{R}^d$ satisfies:*

(i) **Lipschitz in x .** *There exists $L > 0$ such that for all x, y and all η ,*

$$\|f_{\mathcal{D}}(x, \eta) - f_{\mathcal{D}}(y, \eta)\| \leq L\|x - y\|.$$

(ii) **Linear growth.** *There exists $K > 0$ such that for all x, η ,*

$$\|f_{\mathcal{D}}(x, \eta)\| \leq K(1 + \|x\|).$$

(iii) **Measurability.** *For each $\mathcal{D}$, the map $(t, \omega) \mapsto f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}(\omega), \eta_{\kappa(t)}(\omega))$ is progressively measurable with respect to the filtration of (W_t, η_t) .*

Proposition A.6 (Existence and uniqueness). *Suppose Assumption A.5 holds and that $\kappa : [0, \infty) \rightarrow [0, \infty)$ is left-continuous, piecewise constant with finitely many jumps on bounded intervals, and satisfies $\kappa(t) \leq t$. Then for each dataset $\mathcal{D}$ there exists a unique strong solution $(X_t^{\mathcal{D}})_{t \geq 0}$ of*

$$dX_t^{\mathcal{D}} = f_{\mathcal{D}}(X_{\kappa(t)}^{\mathcal{D}}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t$$

with initial condition $X_0^{\mathcal{D}} = x_0$.

Proof sketch. The drift is piecewise constant in time and globally Lipschitz in x with linear growth, so on each interval where κ and η are constant the equation reduces to a common SDE with globally Lipschitz drift. Existence and uniqueness on each such interval follows from standard SDE theory (see, e.g., Theorem 5.2.1 in Oksendal, 2003). Concatenating these solutions across the countably many intervals defined by the jump times of κ yields a unique global strong solution. A more general treatment of SDEs with delayed drift can be found in Zhang (2005). $\square$

A.5 Girsanov's theorem

We now state a standard version of Girsanov's theorem and indicate how it reduces to Theorem 2.12.

Theorem A.7 (Girsanov). *Let $(W_t)_{t \geq 0}$ be a d -dimensional Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, and let $(u_t)_{t \geq 0}$ be an $(\mathcal{F}_t)$ -progressively measurable process with values in $\mathbb{R}^d$. Suppose that*

$$\mathbb{E}_{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T \|u_t\|^2 dt \right) \right] < \infty$$

for some $T > 0$ (Novikov's condition). Define

$$Z_T := \exp \left(- \int_0^T \langle u_t, dW_t \rangle - \frac{1}{2} \int_0^T \|u_t\|^2 dt \right).$$

Then $(Z_t)_{t \in [0, T]}$ is a martingale with $\mathbb{E}_{\mathbb{P}}[Z_T] = 1$, and the measure $\mathbb{Q}$ defined by $d\mathbb{Q} = Z_T d\mathbb{P}$ on $\mathcal{F}_T$ is a probability measure under which

$$\widetilde{W}_t := W_t + \int_0^t u_s ds, \quad 0 \leq t \leq T,$$

is a d -dimensional Brownian motion.

Proof. See Section 3.5 in Karatzkas and Shreve (1991). $\square$

Proof of Theorem 2.12. Apply Theorem A.7 on W_t and $\tilde{u}_t := \sqrt{\beta/2} u_t$ to get $\widetilde{W}_t$. Then under $\mathbb{Q}$ we can write $dX_t^1 = g(X_{\kappa(t)}^1, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} d\widetilde{W}_t$, which has the same law as dX_t^2 under $\mathbb{P}$. $\square$

A.6 Sub-Gaussianity of bounded stochastic integrals

Lemma A.8 (Sub-Gaussian stochastic integral). *Let $(W_t)_{t \geq 0}$ be a d -dimensional BM and $(u_t)_{t \in [0, T]}$ an $\mathcal{F}_t$ -progressively measurable process with $\int_0^T \|u_t\|^2 dt < \infty$ almost surely. Define*

$$M_T := \int_0^T \langle u_t, dW_t \rangle.$$

Then, conditional on the path of (u_t) , M_T is Gaussian with mean 0 and variance $\int_0^T \|u_t\|^2 dt$. Moreover, if $\|u_t\| \leq U$ almost surely for all $t \in [0, T]$, then

$$\mathbb{E}[e^{\lambda M_T}] \leq \exp \left(\frac{\lambda^2 U^2 T}{2} \right),$$

so $M_T \sim \text{subG}(U^2 T)$ in the sense of Definition 2.15.

In the proof of Proposition 2.13 below we apply Lemma A.8 with a uniform bound on $\|u_t\|$ derived from the coupling and drift assumptions to get our desired upper bound.

Proof. To prove the first claim, note that for fixed (u_t) the integral M_T is an L^2 limit of sums

$$\sum_k \langle u_{t_k}, W_{t_{k+1}} - W_{t_k} \rangle,$$

which are Gaussian with mean 0 and variance $\sum_k \|u_{t_k}\|^2 (t_{k+1} - t_k)$; passing to the limit yields variance $\int_0^T \|u_t\|^2 dt$. To prove the second claim, note that

$$\mathbb{E}[e^{\lambda M_T} \mid (u_t)] = \exp\left(\frac{\lambda^2}{2} \int_0^T \|u_t\|^2 dt\right).$$

Taking expectations and using the tower law and $\int_0^T \|u_t\|^2 dt \leq U^2 T$ yield

$$\mathbb{E}[e^{\lambda M_T}] \leq \mathbb{E}\left[\exp\left(\frac{\lambda^2}{2} \int_0^T \|u_t\|^2 dt\right)\right] \leq \exp\left(\frac{\lambda^2 U^2 T}{2}\right),$$

as claimed. $\square$

B Our contributions

B.1 Technical details for the centralized setting

We give the detailed proof of Theorem 3.2. We also record a Bernstein-type refinement of the concentration step that slightly enlarges the admissible range of the Rényi order α .

B.1.1 A Girsanov-based reduction to RDP

We now formalize the reduction from a moment bound on the drift perturbation u_t to a Rényi DP guarantee for the final-state mechanism. We work in the setting of Section 3.2.

Fix adjacent datasets $\mathcal{D}, \mathcal{D}'$, and let

$$(X_t^{\mathcal{D}})_{t \in [0, T]}, \quad (X_t^{\mathcal{D}'})_{t \in [0, T]}$$

be coupled diffusions driven by the same Brownian motion and auxiliary process, as in Section 3.2.2. Let $(Z_t)_{t \in [0, T]}$ be the perturbed diffusion constructed there, so that

$$Z_t = X_t^{\mathcal{D}} \text{ for } t \leq T-1, \quad Z_T \stackrel{a.s.}{=} X_T^{\mathcal{D}'},$$

and let u_t denote the perturbation drift supported on $[T-1, T]$.

By Girsanov's theorem (Theorem 2.12), there exists a probability measure $\mathbb{Q}$ such that $(X_t^{\mathcal{D}})_{t \in [0, T]}$ under $\mathbb{Q}$ has the same law as $(Z_t)_{t \in [0, T]}$ under $\mathbb{P}$, and

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(\sqrt{\frac{\beta}{2}} \int_0^T \langle u_s, dW_s \rangle - \frac{\beta}{4} \int_0^T \|u_s\|^2 ds\right).$$

In particular, $\mathbb{Q}_{\mathcal{A}_{\text{final}}(\mathcal{D})}$ coincides with $\mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')}.$

The next lemma shows that a bound on the MGF of $\int_{T-1}^T \|u_t\|^2 dt$ yields a bound on the Rényi divergence between the output distributions.

Lemma B.1 (RDP via moment bounds on the drift). *Suppose that for all adjacent datasets $\mathcal{D}, \mathcal{D}'$ there exists a perturbation u_t and a measure $\mathbb{Q}$ as above, and that there is a function $\Psi : [0, \infty) \rightarrow [0, \infty)$ such that for all $\theta \geq 0$,*

$$\mathbb{E}_{\mathbb{P}}\left[\exp\left(\theta \int_{T-1}^T \|u_t\|^2 dt\right)\right] \leq \exp(\Psi(\theta)).$$

Then for any $\alpha > 1$,

$$D_{\alpha}(\mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')} \parallel \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D})}) \leq \frac{1}{\alpha - 1} \Psi\left(\frac{\beta}{4} \alpha(\alpha - 1)\right).$$

Proof. By construction, under $\mathbb{Q}$ the law of $X^{\mathcal{D}}$ coincides with that of Z under $\mathbb{P}$, and $Z_T = X_T^{\mathcal{D}'}$ almost surely. Thus $\mathbb{Q}_{\mathcal{A}_{\text{final}}(\mathcal{D})} = \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')}$. Fix $\alpha > 1$. Writing $M_T := \int_0^T \langle u_s, dW_s \rangle$, by Girsanov's,

$$\left(\frac{d\mathbb{Q}}{d\mathbb{P}}\right)^\alpha = \exp\left(\alpha\sqrt{\frac{\beta}{2}} M_T - \frac{\alpha\beta}{4} \int_0^T \|u_s\|^2 ds\right).$$

Under $\mathbb{P}$, M_T is a continuous martingale with quadratic variation $\langle M \rangle_T = \int_0^T \|u_s\|^2 ds$. The exponential martingale identity (cf. 5.12 of Karatzkas and Shreve (2014)) states that for any real λ ,

$$\exp\left(\lambda M_T - \frac{\lambda^2}{2} \langle M \rangle_T\right)$$

is a martingale with expectation 1 under $\mathbb{P}$.

Taking $\lambda = \alpha\sqrt{\beta/2}$, we obtain

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}\left[\left(\frac{d\mathbb{Q}}{d\mathbb{P}}\right)^\alpha\right] &= \mathbb{E}_{\mathbb{P}}\left[\exp\left(\alpha\sqrt{\frac{\beta}{2}} M_T - \frac{\alpha\beta}{4} \int_0^T \|u_s\|^2 ds\right)\right] \\ &= \mathbb{E}_{\mathbb{P}}\left[\exp\left(\alpha\sqrt{\frac{\beta}{2}} M_T - \frac{\alpha^2\beta}{4} \int_0^T \|u_s\|^2 ds\right) \exp\left(\frac{\beta}{4}\alpha(\alpha-1) \int_0^T \|u_s\|^2 ds\right)\right] \\ &= \mathbb{E}_{\mathbb{P}}\left[\exp\left(\frac{\beta}{4}\alpha(\alpha-1) \int_0^T \|u_s\|^2 ds\right)\right], \\ &= \mathbb{E}_{\mathbb{P}}\left[\exp\left(\frac{\beta}{4}\alpha(\alpha-1) \int_{T-1}^T \|u_s\|^2 ds\right)\right] \\ &\leq \exp\left(\Psi\left(\frac{\beta}{4}\alpha(\alpha-1)\right)\right) \end{aligned}$$

because the first exponential factor has expectation 1 and $u_t = 0$ outside $[T-1, T]$, so

$$D_\alpha(\mathbb{Q} \parallel \mathbb{P}) \leq \frac{1}{\alpha-1} \Psi\left(\frac{\beta}{4}\alpha(\alpha-1)\right).$$

Finally, by the data processing inequality and the fact that $\mathbb{Q}_{\mathcal{A}_{\text{final}}(\mathcal{D})} = \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')}$,

$$D_\alpha(\mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D}')} \parallel \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D})}) = D_\alpha(\mathbb{Q}_{\mathcal{A}_{\text{final}}(\mathcal{D})} \parallel \mathbb{P}_{\mathcal{A}_{\text{final}}(\mathcal{D})}) \leq D_\alpha(\mathbb{Q} \parallel \mathbb{P}),$$

which yields the claimed bound. $\square$

B.1.2 Proof of Theorem 3.2

Proof. We now prove Theorem 3.2. Throughout this subsection we assume Assumption 3.1 holds, and we work with two adjacent datasets $\mathcal{D}, \mathcal{D}'$ differing only in their j th datum.

As in Section 3, let $(x_n^{\mathcal{D}})_{n \geq 0}$ and $(x_n^{\mathcal{D}'})_{n \geq 0}$ denote the SGLD iterates started from the same $x_0 \in \mathbb{R}^d$, with shared Gaussian noise and minibatches $(A_{n+1})_{n \geq 0}$. Let $(X_t^{\tilde{\mathcal{D}}})_{t \geq 0}$, $\tilde{\mathcal{D}} \in \{\mathcal{D}, \mathcal{D}'\}$, be the continuous-time interpolation solving

$$dX_t^{\tilde{\mathcal{D}}} = f_{\tilde{\mathcal{D}}}(X_{\kappa(t)}^{\tilde{\mathcal{D}}}, \eta_{\kappa(t)}) dt + \sqrt{2/\beta} dW_t,$$

where $\kappa(t) = \lfloor t \rfloor$, $\eta_n := A_{n+1}$, and

$$f_{\mathcal{D}}(x, \eta) := -\gamma \nabla k(x) - \frac{\gamma}{s} \sum_{i \in \eta} \nabla_x v(x, d_i).$$

By construction, $(X_n^{\mathcal{D}})_{n \in \mathcal{N}}$ and $(x_n^{\mathcal{D}'})_{n \in \mathcal{N}}$ have the same law.

Following Bertazzi et al., we construct a perturbed process $(Z_t)_{t \in [0, T]}$ by

$$dZ_t = f_{\mathcal{D}}(Z_{\kappa(t)}, \eta_{\kappa(t)}) dt + u_t dt + \sqrt{2/\beta} dW_t, \quad Z_0 = x_0,$$

with perturbation

$$u_t := \mathbf{1}_{\{t \in [T-1, T]\}} \left[f_{\mathcal{D}'}(X_{\kappa(t)}^{\mathcal{D}'}, \eta_{\kappa(t)}) - f_{\mathcal{D}}(Z_{\kappa(t)}, \eta_{\kappa(t)}) - (Z_{T-1} - X_{T-1}^{\mathcal{D}'}) \right].$$

As in Proposition 4.5 of Bertazzi et al., this choice guarantees that

$$Z_t = X_t^{\mathcal{D}} \quad \text{for } t \leq T-1, \quad Z_T = X_T^{\mathcal{D}'}$$

almost surely. In particular, the law of Z_T under $\mathbb{P}$ coincides with the law of $X_T^{\mathcal{D}'}$ under $\mathbb{P}$.

Our strategy is to verify the MGF condition in Lemma B.1 with an explicit function Ψ , and to read off the resulting RDP parameters.

On $[T-1, T]$, we have $\kappa(t) = T-1$, so u_t is constant there; write $u := u_t$ for $t \in [T-1, T]$ and define $e_n := X_n^{\mathcal{D}} - X_n^{\mathcal{D}'}$. Since $Z_{T-1} = X_{T-1}^{\mathcal{D}}$, on $t \in [T-1, T]$,

$$\begin{aligned} \|u\| &\leq (\gamma L_k + \gamma L_v + 1) \|e_{T-1}\| + \|f_{\mathcal{D}'}(Z_{\kappa(t)}, \eta_{\kappa(t)}) - f_{\mathcal{D}}(Z_{\kappa(t)}, \eta_{\kappa(t)})\| \\ &= (\gamma L_k + \gamma L_v + 1) \|e_{T-1}\| + \frac{\gamma}{s} \mathbf{1}_{\{j \in \eta_{\kappa(t)}\}} \|\nabla_x v(x, d_j) - \nabla_x v(x, d'_j)\| \\ &\leq (\gamma L_k + \gamma L_v + 1) \|e_{T-1}\| + \frac{\gamma \Delta_v}{s} \mathbf{1}_{\{j \in A_T\}} \end{aligned}$$

where we used the Lipschitzness of ∇k and $\nabla_x v$ in x to bound the first term, and the Δ_v -sensitivity of $\nabla_x v$ in the data component to bound the second. Writing $A := (1 + \gamma(L_k + L_v))$ and $b := \gamma \Delta_v / s$,

$$\|u\| \leq A \|e_{T-1}\| + b \mathbf{1}_{\{j \in A_T\}}.$$

Because the A_T are drawn independently of the past, i.e. the minibatches are sampled i.i.d. over time, the indicator $\mathbf{1}_{\{j \in A_T\}}$ is independent of e_{T-1} . Hence, for any $\theta \geq 0$,

$$\begin{aligned} \mathbb{E} \left[\exp \left(\theta \int_{T-1}^T \|u_t\|^2 dt \right) \right] &\leq \mathbb{E}[\exp(2\theta A^2 \|e_{T-1}\|^2)] \mathbb{E}[\exp(2\theta b^2 \mathbf{1}_{\{j \in A_T\}})], \\ &= \mathbb{E}[\exp(2\theta A^2 \|e_{T-1}\|^2)] [(1-q) + qe^{2\theta b^2}] \end{aligned}$$

where we use $(a+b)^2 \leq 2a^2 + 2b^2$ on $\|u_t\|^2$ and $\mathbf{1}_{\{j \in A_T\}} \sim \text{Bern}(q)$. To apply Lemma B.1 and get our guarantee, it remains to bound the left term. Write $T(x) := x - \gamma \nabla k(x)$ and observe that

$$e_{n+1} \leq T(X_n^{\mathcal{D}}) - T(X_n^{\mathcal{D}'}) - \frac{\gamma}{s} \Delta_v \mathbf{1}_{\{j \in A_{n+1}\}}$$

again by the Δ_v -sensitivity of $\nabla_x v$ and our definition of adjacency. Using the strong convexity and Lipschitzness of ∇k , we obtain the standard contraction bound

$$\begin{aligned} \|T(x) - T(y)\|^2 &\leq \|x - y\|^2 - 2\gamma \langle \nabla k(x) - \nabla k(y), x - y \rangle + \gamma^2 \|\nabla k(x) - \nabla k(y)\|^2 \\ &\leq (1 - 2\gamma a + \gamma^2 L_k^2) \|x - y\|^2, \end{aligned}$$

where $\rho := 1 - 2\gamma a + \gamma^2 L_k^2 \in (0, 1)$ because we assumed our step size $\gamma \in (0, 2a/L_k^2)$. As such

$$\|e_{n+1}\| \leq \rho \|e_n\| + b \mathbf{1}_{\{j \in A_{n+1}\}}.$$

Define an auxiliary process $\tilde{e}_n$ by $\tilde{e}_{n+1} = \rho \tilde{e}_n + b \mathbf{1}_{\{j \in A_{n+1}\}}$ with $\tilde{e}_0 = 0$, so that we get $\|e_n\| \leq \tilde{e}_n$. The idea is that now we just have to bound $\mathbb{E}[\exp(\theta \tilde{e}_n^2)]$. Unrolling the recursion, we see that

$$\tilde{e}_n = b \sum_{k=1}^n \rho^{n-k} \mathbf{1}_{\{j \in A_k\}}.$$

Write

$$S_n := \sum_{k=1}^n c_k (\mathbf{1}_{\{j \in A_k\}} - q), \quad c_k := \rho^{n-k}.$$

Since the $c_k(\mathbf{1}_{\{j \in A_k\}} - q)$'s are mean-zero, independent, and a.s. bounded in $[-c_k q, c_k(1-q)]$, we can apply Lemma 2.16 term-by-term to get that

$$\mathbb{E}[\exp(s S_n)] \leq \exp \left(\frac{s^2}{8} \sum_{k=1}^n c_k^2 \right) \leq \exp \left(\frac{s^2}{8(1-\rho^2)} \right)$$

with the second inequality because $\rho \in (0, 1)$. This implies $S_n \sim \text{subG}\left(\frac{1}{4(1-\rho^2)}\right)$. As such

$$Y := \tilde{e}_n - \mathbb{E}\tilde{e}_n = bS_n \sim \text{subG}(\sigma_e^2)$$

with $\sigma_e^2 \leq \frac{b^2}{4(1-\rho^2)}$. By Lemma 2.18, this means

$$E := (\tilde{e}_n - \mathbb{E}\tilde{e}_n)^2 - \mathbb{E}(\tilde{e}_n - \mathbb{E}\tilde{e}_n)^2 \sim \text{subE}(16\sigma_e^2).$$

We want to write $\tilde{e}_n^2$ in terms of Y and E . Since $(\mathbb{E}\tilde{e}_n)^2 = \mathbb{E}\tilde{e}_n^2 - \mathbb{E}(\tilde{e}_n - \mathbb{E}\tilde{e}_n)^2$, actually

$$\tilde{e}_n^2 = (\tilde{e}_n - \mathbb{E}\tilde{e}_n)^2 + 2\mathbb{E}\tilde{e}_n(\tilde{e}_n - \mathbb{E}\tilde{e}_n) + (\mathbb{E}\tilde{e}_n)^2 = \mathbb{E}\tilde{e}_n^2 + 2(\mathbb{E}\tilde{e}_n)Y + E.$$

By Cauchy-Schwarz and our Y and E being sub-Gaussian and sub-exponential respectively,

$$\mathbb{E}[\exp(\lambda\tilde{e}_n^2)] \leq e^{\lambda\mathbb{E}\tilde{e}_n^2}(\mathbb{E}[e^{2\lambda E}])^{1/2}(\mathbb{E}[e^{4\lambda(\mathbb{E}\tilde{e}_n)Y}])^{1/2} \leq e^{\lambda\mathbb{E}\tilde{e}_n^2}e^{256\sigma_e^4\lambda^2}e^{4\lambda(\mathbb{E}\tilde{e}_n)Y}$$

when $|\lambda| \leq \frac{1}{32\sigma_e^2} \leq \frac{1-\rho^2}{8b^2}$. Since $\mathbb{E}\tilde{e}_n \leq \frac{bq}{1-\rho}$ and $\mathbb{E}\tilde{e}_n^2 \leq b^2\left(\frac{q}{1-\rho^2} + \frac{q^2}{(1-\rho)^2}\right)$ because $\rho \in (0, 1)$, we can simplify the right hand side to

$$\mathbb{E}[\exp(\lambda\tilde{e}_n^2)] \leq \exp(L_1\lambda + L_2\lambda^2)$$

with $L_1 := b^2\left(\frac{q}{1-\rho^2} + \frac{q^2}{(1-\rho)^2}\right)$ and $L_2 := \frac{17b^4}{(1-\rho)^2(1-\rho^2)}$. To get rid of a quadratic dependence on λ in the upper bound, we take $\lambda \leq \frac{q(1-\rho)^2}{17b^2}$ so that $L_2\lambda^2 \leq L_1\lambda$, and thus on $\lambda \leq \frac{q(1-\rho)^2}{17b^2}$,

$$\mathbb{E}[\exp(\lambda\|\tilde{e}_n\|^2)] \leq \mathbb{E}[\exp(\lambda\tilde{e}_n^2)] \leq \exp(2L_1\lambda).$$

Putting it all together, we conclude that when $|2\theta A^2| \leq \frac{q(1-\rho)^2}{17b^2}$, i.e. when $|\theta| \leq \frac{q(1-\rho)^2}{34A^2b^2}$

$$\mathbb{E}\left[\exp\left(\theta \int_{T-1}^T \|u_t\|^2 dt\right)\right] \leq \exp(4L_1A^2\theta)[(1-q) + qe^{2\theta b^2}].$$

Thus the MGF condition of Lemma B.1 holds with

$$\Psi(\theta) = 4L_1A^2\theta + \log[(1-q) + qe^{2\theta b^2}],$$

provided $\theta \leq q(1-\rho^2)/(34A^2b^2)$. At this point applying Lemma B.1 and plugging in $\theta = \frac{\beta}{4}\alpha(\alpha-1)$ to $\Psi(\theta)$ gives us our desired result. $\square$

B.1.3 Bernstein-based refinement of the coupling error

In this subsection we refine the analysis of the coupling error process in the proof of Theorem 3.2 by replacing the Hoeffding-based control of the weighted Bernoulli sum

$$S_n := \sum_{k=1}^n c_k (\mathbf{1}_{\{j \in A_k\}} - q), \quad c_k := \rho^{n-k},$$

with a Bernstein-type MGF bound that takes into account the variance $q(1-q)$ of the Bernoulli indicators. This yields a sharper sub-Gaussian proxy for S_n and, consequently, a slightly larger admissible range for the Rényi order α in Theorem 3.2. The structure of the proof remains identical to Appendix B.1.2; we only spell out the portions that change.

B.1.4 A Bernstein MGF bound for bounded independent variables

We begin with a standard Bernstein-type moment generating function (MGF) bound for sums of independent, bounded, mean-zero random variables.

Lemma B.2 (Bernstein MGF bound). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_k = 0$, $|X_k| \leq B$ almost surely, and $\text{Var}(X_k) \leq \sigma_k^2$. Let*

$$V_n := \sum_{k=1}^n \sigma_k^2.$$

Then for all $s \in \mathbb{R}$ with $|s| \leq \frac{1}{2B}$,

$$\mathbb{E}\left[\exp\left(s \sum_{k=1}^n X_k\right)\right] \leq \exp(2s^2 V_n).$$

In particular, $\sum_{k=1}^n X_k$ is sub-Gaussian with variance proxy $4V_n$ on the range $|s| \leq \frac{1}{2B}$.

Proof. For each k , Taylor expanding e^{sX_k} and using $|X_k| \leq B$ gives

$$\mathbb{E}[e^{sX_k}] = 1 + \frac{s^2}{2}\mathbb{E}[X_k^2] + R_k(s),$$

with the remainder $|R_k(s)| \leq \frac{|s|^3}{6}\mathbb{E}[|X_k|^3] \leq \frac{|s|^3 B}{6}\mathbb{E}[X_k^2]$. For $|s| \leq 1/(2B)$, we have $|s|B \leq \frac{1}{2}$, so

$$|R_k(s)| \leq \frac{|s|B}{3} \cdot \frac{s^2}{2}\mathbb{E}[X_k^2] \leq \frac{1}{6} \cdot \frac{s^2}{2}\mathbb{E}[X_k^2].$$

Thus

$$\mathbb{E}[e^{sX_k}] \leq 1 + \frac{s^2}{2}\mathbb{E}[X_k^2]\left(1 + \frac{1}{6}\right) \leq \exp\left(\frac{7}{12}s^2\mathbb{E}[X_k^2]\right) \leq \exp\left(\frac{7}{12}s^2\sigma_k^2\right),$$

where we use the identity $1 + x \leq e^x$ for $x \geq 0$. Since the X_k are independent,

$$\mathbb{E}\left[\exp\left(s \sum_{k=1}^n X_k\right)\right] = \prod_{k=1}^n \mathbb{E}[e^{sX_k}] \leq \exp\left(\frac{7}{12}s^2 \sum_{k=1}^n \sigma_k^2\right) \leq \exp(2s^2 V_n),$$

as desired. $\square$

B.1.5 Application to the weighted Bernoulli sum S_n

We now apply Lemma B.2 to

$$S_n = \sum_{k=1}^n c_k(\mathbf{1}_{\{j \in A_k\}} - q), \quad c_k := \rho^{n-k}.$$

Recall from Appendix B.1.2 that the indicators $\mathbf{1}_{\{j \in A_k\}}$ are independent $\text{Bern}(q)$ r.v.s. Define

$$X_k := c_k(\mathbf{1}_{\{j \in A_k\}} - q).$$

Then $\mathbb{E}[X_k] = 0$, $|X_k| \leq |c_k| \leq 1$, and

$$\text{Var}(X_k) = c_k^2 \text{Var}(\mathbf{1}_{\{j \in A_k\}}) = q(1-q)c_k^2.$$

Thus in Lemma B.2 we may take $B = 1$ and

$$V_n = \sum_{k=1}^n \text{Var}(X_k) = q(1-q) \sum_{k=1}^n c_k^2 \leq q(1-q) \sum_{k=0}^{\infty} \rho^{2k} = \frac{q(1-q)}{1-\rho^2}.$$

For all s with $|s| \leq 1/2$, Lemma B.2 gives

$$\mathbb{E}[e^{sS_n}] \leq \exp(2s^2 V_n) \leq \exp\left(2s^2 \frac{q(1-q)}{1-\rho^2}\right).$$

In other words, on $|s| \leq 1/2$,

$$S_n \sim \text{subG}(\sigma_S^2), \quad \sigma_S^2 := \frac{4q(1-q)}{1-\rho^2},$$

since $\exp\left(2s^2 \frac{q(1-q)}{1-\rho^2}\right) = \exp\left(\frac{\sigma_S^2 s^2}{2}\right)$. In particular, writing $b := \frac{\gamma \Delta_v}{s}$,

$$Y := \tilde{e}_n - \mathbb{E}\tilde{e}_n = bS_n, \sim \text{subG}(\sigma_e^2), \quad \sigma_e^2 \leq b^2 \sigma_S^2 = \frac{4b^2 q(1-q)}{1-\rho^2}.$$

This refines the Hoeffding-based bound in B.1.2, which had $\sigma_e^2 \leq \frac{b^2}{4(1-\rho^2)}$ independent of q .

B.2 Proof of Theorem 4.1

Proof. Our proof follows that of Theorem 3.2. For datasets $\mathcal{D}$ and $\mathcal{D}'$ differing in the i th hospital at the j th datum, we can define $b_i := \gamma \frac{N_i}{f_i s} \Delta_v$ (In DSGLD our b_i differs from the centralized SGLD quantity $b := \gamma \Delta_v / s$ because the local gradient at client i is rescaled by the factor $\frac{N_i}{f_i m_{\text{loc}}}$, accounting for client subsampling with probability f_i and a local minibatch of size m_{loc} out of N_i points, to obtain an unbiased estimator of the global gradient, and this rescaling multiplies the corresponding drift difference and hence the sensitivity by the same factor). Denote our DSGLD iterates $(x_n^{\hat{\mathcal{D}}})_{n \geq 0}$ with $x_0 \in \mathbb{R}^d$. Specific to the i th hospital we can consider the continuous-time interpolation of the DSGLD algorithm $(X_t^{\hat{\mathcal{D}}})_{t \geq 0}$

$$dX_t^{\hat{\mathcal{D}}} = f_{\hat{\mathcal{D}}}(X_{\kappa(t)}^{\hat{\mathcal{D}}}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) dt + \sqrt{\frac{2}{\beta}} dW_t$$

for $\hat{\mathcal{D}}, \kappa, \eta$ defined as in Theorem 3.2 and with scheduler $\iota_{\kappa(t)}$, i.e. at iteration t we sample a batch from hospital ι_t , giving for batches $\eta_{\kappa(t)}$

$$f_{\mathcal{D}}(x, i, B) := -\gamma \nabla k(x) - \gamma \frac{N_i}{f_i s} \sum_{j \in B} \nabla_x v(x, d_j)$$

we can construct the perturbed process $(Z_t^i)_{t \in [0, T]}$ as

$$dZ_t^i = f_{\mathcal{D}}(Z_{\kappa(t)}^i, \iota_{\kappa(t)}, \eta_{\kappa(t)}) dt + u_t dt + \sqrt{\frac{2}{\beta}} dW_t$$

with perturbation

$$u_t := \mathbf{1}_{\{t \in [T-1, T]\}} \left[f_{\mathcal{D}'}(X_{\kappa(t)}^{\mathcal{D}'}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) - f_{\mathcal{D}}(Z_{\kappa(t)}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) - (Z_{T-1} - X_{T-1}^{\mathcal{D}'}) \right]$$

write u and e_T as in Theorem 3.2. For $t \in [T-1, T]$ we now have for $A := 1 + \gamma(L_k + L_v)$, $B_{T-1}(i) := \mathbf{1}_{\{\iota_{T-1}=i\}} \mathbf{1}_{\{j \in \eta_{T-1}\}}$ that

$$\begin{aligned} \|u\| &\leq (1 + \gamma(L_k + L_v)) \|e_{T-1}\| + \|f_{\mathcal{D}'}(Z_{T-1}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1})\| \\ &\leq (1 + \gamma(L_k + L_v)) \|e_{T-1}\| + \gamma \frac{N_i}{f_i s} \Delta_v B_{T-1}(i) \end{aligned}$$

note that $B_{T-1}(i)$ is independent of e_{T-1} . As we maintain the same assumptions of $\gamma \in (0, 2a/L_k^2)$, convexity and Lipschitzness of ∇k , we have

$$\|e_{n+1}^i\| \leq \rho \|e_n^i\| + b_i B_{T-1}(i)$$

then we again arrive at a recursive relation

$$\tilde{e}_n = b_i \sum_{k=1}^n \rho^{n-k} B_{T-1}(i)$$

we can now proceed precisely as in Theorem 3.2. We find $E := Y^2 - \mathbb{E}Y^2 \sim \text{subE}\left(\frac{4b_i^2}{1-\rho^2}\right)$ and again with $\theta \leq q_i(1-\rho^2)/(34A^2b_i^2)$ may apply Lemma B.1 and take $\theta = \frac{\beta}{4}\alpha(\alpha-1)$. To reach our desired result, we concede the maximum loss over all hospitals $\mathcal{D}^{(1)}, \dots, \mathcal{D}^{(S)}$. $\square$

B.3 Proof of Theorem 5.2

Proof. Our proof follows that of Theorem 3.2 and Theorem 4.1. For datasets $\mathcal{D}$ and $\mathcal{D}'$ differing in the i th hospital at the j th datum, take $b_i, \hat{\mathcal{D}}$, and $\iota_{\kappa(t)}$ as defined prior.

Our continuous-time interpolation $(X_t^{\hat{\mathcal{D}}})_{t \geq 0}$ now solves

$$dX_t^{\hat{\mathcal{D}}} = f_{\hat{\mathcal{D}}}^{\text{FS}}(X_{\kappa(t)}^{\hat{\mathcal{D}}}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) dt + \sqrt{\frac{2}{\beta}} dW_t,$$

where we have

$$f_{\mathcal{D}}^{\text{FS}}(x, i, B) := -\gamma \nabla k(x) - \gamma \frac{N_i}{f_{iS}} \sum_{j \in B} \nabla_x v(x, d_j) + \gamma g_{i, \mathcal{D}}(x),$$

with $g_{i, \mathcal{D}}$ the hospital-specific conducive gradients from Assumption 5.1.

We construct a perturbed process $(Z_t^i)_{t \in [0, T]}$ by

$$dZ_t^i = f_{\mathcal{D}}^{\text{FS}}(Z_{\kappa(t)}^i, \iota_{\kappa(t)}, \eta_{\kappa(t)}) dt + u_t dt + \sqrt{\frac{2}{\beta}} dW_t,$$

with perturbation

$$u_t := \mathbf{1}_{\{t \in [T-1, T]\}} \left[f_{\mathcal{D}'}^{\text{FS}}(X_{\kappa(t)}^{\mathcal{D}'}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) - f_{\mathcal{D}}^{\text{FS}}(Z_{\kappa(t)}, \iota_{\kappa(t)}, \eta_{\kappa(t)}) - (Z_{T-1} - X_{T-1}^{\mathcal{D}'}) \right],$$

so that we have $Z_t^i = X_t^{\mathcal{D}}$ for $t \leq T-1$ and $Z_T^i = X_T^{\mathcal{D}'}$ almost surely. We again write $u := u_t$ and $e_n^i := X_n^{\mathcal{D}} - X_n^{\mathcal{D}'}$.

On the interval $[T-1, T]$, we have $\kappa(t) = T-1$, hence

$$\begin{aligned} \|u\| &= \|f_{\mathcal{D}'}^{\text{FS}}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}^{\text{FS}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1}) - (Z_{T-1} - X_{T-1}^{\mathcal{D}'})\| \\ &\leq \|Z_{T-1} - X_{T-1}^{\mathcal{D}'}\| + \|f_{\mathcal{D}'}^{\text{FS}}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}^{\text{FS}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1})\| \end{aligned}$$

note that

$$\begin{aligned} &\|f_{\mathcal{D}'}^{\text{FS}}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}^{\text{FS}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1})\| \\ &\leq \|f_{\mathcal{D}'}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1})\| \\ &\quad + \gamma \|g_{\iota_{T-1}, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'}) - g_{\iota_{T-1}, \mathcal{D}}(Z_{T-1})\| \end{aligned}$$

for the left term, the same argument as in Theorem 4.1 yields

$$\|f_{\mathcal{D}'}(X_{T-1}^{\mathcal{D}'}, \iota_{T-1}, \eta_{T-1}) - f_{\mathcal{D}}(Z_{T-1}, \iota_{T-1}, \eta_{T-1})\| \leq \gamma(L_k + L_v) \|e_{T-1}^i\| + b_i B_{T-1}(i),$$

where

$$B_{T-1}(i) := \mathbf{1}_{\{\iota_{T-1}=i\}} \mathbf{1}_{\{j \in \eta_{T-1}\}}$$

for the latter term, using Assumption 5.1 we have

$$\begin{aligned} \|g_{i, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'} - g_{i, \mathcal{D}}(Z_{T-1}))\| &\leq \|g_{i, \mathcal{D}'}(X_{T-1}^{\mathcal{D}'} - g_{i, \mathcal{D}'}(Z_{T-1}))\| + \|g_{i, \mathcal{D}'}(Z_{T-1}) - g_{i, \mathcal{D}}(Z_{T-1})\| \\ &\leq L_g \|e_{T-1}^i\| + \Delta_i^{(g)}, \end{aligned}$$

where $\Delta_i^{(g)} = (f_i^{-1} - 1) \Delta_i^{(q)}$ follows from the definition of $g_{i, \mathcal{D}}$ and the fact that only the i th hospital changes between $\mathcal{D}$ and $\mathcal{D}'$. Then

$$\begin{aligned} \|u\| &\leq (1 + \gamma(L_k + L_v + L_g)) \|e_{T-1}^i\| + b_i B_{T-1}(i) + \gamma \Delta_i^{(g)} \\ &= A^{\text{FS}} \|e_{T-1}^i\| + b_i B_{T-1}(i) + c_i \end{aligned}$$

Using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$, we have

$$\|u\|^2 \leq 3(A^{\text{FS}})^2 \|e_{T-1}^i\|^2 + 3b_i^2 B_{T-1}(i)^2 + 3c_i^2$$

Thus, for any $\theta \geq 0$,

$$\begin{aligned} \mathbb{E} \left[\exp \left(\theta \int_{T-1}^T \|u_t\|^2 dt \right) \right] &= \mathbb{E} [\exp(\theta \|u\|^2)] \\ &\leq \exp(3\theta c_i^2) \mathbb{E} [\exp(3\theta (A^{\text{FS}})^2 \|e_{T-1}^i\|^2)] \mathbb{E} [\exp(3\theta b_i^2 B_{T-1}(i)^2)] \\ &= \exp(3\theta c_i^2) \mathbb{E} [\exp(3\theta (A^{\text{FS}})^2 \|e_{T-1}^i\|^2)] [(1 - q_i) + q_i e^{3\theta b_i^2}], \end{aligned}$$

since $B_{T-1}(i) \sim \text{Bern}(q_i)$ and is independent of e_{T-1}^i .

The recursion for e_n^i is unchanged in its DSGLD part, so by the same argument as Theorem 4.1 we have for

$$\lambda := 3\theta(A^{\text{FS}})^2 \quad \text{and} \quad \lambda \leq \frac{q_i(1-\rho)^2}{17b_i^2}$$

the bound

$$\mathbb{E}[\exp(\lambda \|e_{T-1}^i\|^2)] \leq \exp(2L_1(i)\lambda)$$

with $L_1(i)$ as in the statement. Then under $\theta \leq \frac{q_i(1-\rho)^2}{51(A^{\text{FS}})^2 b_i^2}$ we have $\lambda \leq q_i(1-\rho)^2/(17b_i^2)$, and hence

$$\mathbb{E}[\exp(3\theta(A^{\text{FS}})^2 \|e_{T-1}^i\|^2)] \leq \exp(6L_1(i)(A^{\text{FS}})^2\theta)$$

Thus the MGF condition of Lemma B.1 holds with

$$\Psi_i^{\text{FS}}(\theta) := 6L_1(i)(A^{\text{FS}})^2\theta + 3c_i^2\theta + \log[(1-q_i) + q_ie^{3\theta b_i^2}]$$

provided $\theta \leq q_i(1-\rho^2)/(51(A^{\text{FS}})^2 b_i^2)$.

Applying Lemma B.1 as in Theorem 3.2, taking $\theta = \frac{\beta}{4}\alpha(\alpha-1)$ for Lemma B.1 and conceding the maximum gives the desired bound. $\square$